

Uniform ergodicity and spectral gap of importance-tempered MCMC with heavy-tailed targets

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Abstract

We study the importance-tempered random-walk Metropolis (RWM) algorithm on $\mathbb{R}^d$, which targets the tempered density $\pi_\beta(x) \propto \pi(x)^\beta$ with $\beta \in (0, 1)$ and corrects for the resulting bias by importance weighting. The dynamics of this sampler can be described by a continuous-time Markov chain whose embedded chain is the tempered RWM. We prove that when the tails of π decay at rate $\|x\|^{-\gamma}$, this continuous-time chain is uniformly ergodic if and only if $d/\gamma < \beta < (\gamma - 2)/\gamma$. Further, the global spectral gap of this chain can be controlled via the spectral gap of the chain restricted to a ball with radius growing polynomially in d . Our theory and simulation studies show that, with a suitable choice of β , importance tempering can substantially reduce sensitivity to initialization and significantly improve efficiency over the standard RWM. *Keywords:* continuous-time Markov chain, drift condition, heavy-tailed distribution, Metropolis–Hastings algorithm, self-normalized importance sampling, spectral gap, uniform ergodicity, variance reduction.

1 Introduction

1.1 Importance-tempered random-walk Metropolis algorithms

Let Π be a continuous probability distribution on $\mathbb{R}^d$ with density function $\pi > 0$ everywhere. A standard tool for sampling from Π is the random-walk Metropolis (RWM) algorithm: given current state x , we propose $x' = x + \sigma Z$, where $\sigma > 0$ and Z has a symmetric density κ , and then accept or reject x' according to the Metropolis rule. Geometric ergodicity and uniform ergodicity are desirable convergence properties for such Markov chains. The former requires convergence to Π at an exponential rate, while the latter further demands that this rate be bounded uniformly over all initial states. The seminal work of Mengersen and Tweedie [17] showed that RWM is never uniformly ergodic on $\mathbb{R}^d$. Intuitively, the time needed for the algorithm to enter the high-density region can be made arbitrarily large by starting far in the tails, which prevents uniform convergence over all initial states. Moreover, Mengersen and Tweedie [17] proved that when $d = 1$, RWM with

$\int |z| \kappa(z) < \infty$ is geometrically ergodic if and only if Π has exponentially light tails; when $d > 1$, this exponential decay condition remains necessary but is not sufficient for geometric ergodicity [22, 10].

In this work, we consider a simple approach to achieving uniform ergodicity with heavy-tailed targets. Rather than targeting Π directly, we run RWM with stationary density $\pi_\beta(x) \propto \pi(x)^\beta$ for some $\beta \in (0, 1)$ and correct for the resulting bias by importance weighting; this idea can be traced back at least to Jennison [11]. Formally, let $X_1, \dots, X_n$ denote samples generated from this RWM algorithm and define $\Pi(f) := \int f d\Pi$ for an integrable function f . We estimate $\Pi(f)$ using the self-normalized importance sampling (SNIS) estimator

$$\tilde{\Pi}_{\beta,n}(f) := \frac{\sum_{i=1}^n f(X_i) w(X_i)}{\sum_{i=1}^n w(X_i)}, \text{ where } w(x) = \frac{\pi(x)}{\pi_\beta(x)} \propto \pi(x)^{1-\beta}. \quad (1)$$

The importance weights $w(X_1), \dots, w(X_n)$ only need to be evaluated up to a normalizing constant and can be reused for estimating $\Pi(f_j)$ for any collection of integrable functions $\{f_j\}$. Following Gramacy et al. [7] and Zanella and Roberts [28], we refer to this procedure as importance-tempered RWM and summarize it in Algorithm 1.

Such a Markov chain importance sampling scheme can be extended to more general auxiliary stationary distributions. We focus on the tempered density π_β , since this choice requires only a minimal modification of the standard RWM; indeed, in almost every applied setting, the same MCMC algorithm can be used to sample from π or π_β with same computational cost and implementation effort, except that for tempered targets one needs to keep track of unnormalized importance weights. We will show that this nearly zero-cost modification can substantially improve estimator's precision and robustness in a broad range of settings.

Algorithm 1. Importance-tempered random-walk Metropolis.

Input: target density $\pi(x)$, symmetric proposal density $\kappa(z)$, step size $\sigma > 0$,
inverse temperature $\beta \in (0, 1)$, number of iterations n , initial state $x_0 \in \mathbb{R}^d$.
For $i = 1$ **to** n **do**
 Draw $z_i \sim \kappa(\cdot)$ and set $x^* \leftarrow x_{i-1} + \sigma z_i$.
 Compute the acceptance probability $p_i \leftarrow 1 \wedge \frac{\pi(x^*)^\beta}{\pi(x_{i-1})^\beta}$.
 Draw $u_i \sim \text{Unif}(0, 1)$.
 If $u_i \leq p_i$ **then** set $x_i \leftarrow x^*$; **else** set $x_i \leftarrow x_{i-1}$.
 Compute the unnormalized importance weight $\tilde{w}_i \leftarrow \pi(x_i)^{1-\beta}$.
End for
Output: samples $(x_1, \dots, x_n)$ and weights $(\tilde{w}_1, \dots, \tilde{w}_n)$.

1.2 Main results

The asymptotic variance of the SNIS estimator $\tilde{\Pi}_{\beta,n}(f)$ depends on both the chain's convergence rate and the efficiency of importance weighting. For theoretical analysis, a convenient and fruitful strategy is to approximate $\tilde{\Pi}_{\beta,n}(f)$ by the unweighted time average of a continuous-time Markov chain $(Y_t)_{t \geq 0}$, whose embedded chain is $(X_i)_{i \geq 1}$ and holding time at X_i has mean equal to the importance weight $w(X_i)$; see Section 2.1 for details.

Our first main result identifies the exact regime in which the continuous-time chain $(Y_t)_{t \geq 0}$ is uniformly ergodic. Specifically, we prove that when the tails of Π decay at the polynomial rate $\|x\|^{-\gamma}$, $(Y_t)_{t \geq 0}$ is uniformly ergodic if and only if $d/\gamma < \beta < (\gamma - 2)/\gamma$; see Theorem 1. Clearly, such β exists if and only if $\gamma > d + 2$. When Π has exponentially light tails, $(Y_t)_{t \geq 0}$ is uniformly ergodic for any $\beta \in (0, 1)$; see Corollary 1. This marks a sharp contrast with standard RWM, which is never uniformly ergodic on $\mathbb{R}^d$ and is not even geometrically ergodic for polynomial-tailed targets. Our second result concerns the spectral gap of $(Y_t)_{t \geq 0}$, which we denote by λ . When Π is the t -distribution with $m > 2$ degrees of freedom, we show in Theorem 3 that $\lambda = \Omega(\lambda_R/R^2)$, where λ_R denotes the spectral gap of the chain $(Y_t)_{t \geq 0}$ restricted to the ball $\{x: \|x\| \leq R\}$, and the radius R grows polynomially in d . This reduces the difficult global spectral-gap problem on $\mathbb{R}^d$ to a bounded-domain problem, where more tools are available. Again, standard RWM has fundamentally different behavior: it has no positive spectral gap when the target has polynomial tails on $\mathbb{R}^d$.

For Algorithm 1, the uniform ergodicity of $(Y_t)_{t \geq 0}$ means that the total importance weight accumulated in the tails is uniformly controlled over all initial states, so the performance of the estimator $\tilde{\Pi}_{\beta,n}(f)$ is largely insensitive to the initialization scheme and length of the burn-in period. Section B.1 in the supplement presents a numerical study illustrating this phenomenon. Nevertheless, the number of iterations needed for Algorithm 1 to enter the high-density region can still be unbounded, since $(Y_t)_{t \geq 0}$ can make infinitely many jumps over a fixed time interval. We provide another simulation study in Section B.2 to illustrate that once the sampler enters the high-density region, the accumulated importance weight there quickly dominates the contribution from the tails and significantly reduces the asymptotic variance relative to the standard RWM run for the same number of iterations.

1.3 Related works

Our uniform ergodicity results are closely related to the line of work on sampling algorithms derived from state-dependent Langevin diffusions. Stramer and Tweedie [24] and Roberts and Stramer [21] showed that, even when Π has polynomial tails, one may construct uniformly ergodic Langevin-type diffusions by speeding up the process in the tails. Specifically, our continuous-time chain $(Y_t)_{t \geq 0}$ leaves x at a rate proportional to $\pi(x)^{\beta-1}$ (see Section 2.1). This tail acceleration is the same as in the Langevin tempered diffusion (LTD)

studied by Roberts and Stramer [21], which has stationary distribution Π and diffusion matrix $\pi(x)^{\beta-1}\mathbf{I}$ (with notation modified to facilitate comparison). For polynomial-tailed Π on $\mathbb{R}^d$, Roberts and Stramer [21] showed that LTD is geometrically ergodic if β is chosen appropriately. When $d = 1$, Stramer and Tweedie [24] showed that LTD is uniformly ergodic if $\beta < (\gamma - 2)/\gamma$; the proof is based on verifying a drift condition with Lyapunov function $V(x) = C - (|x| \vee r)^{-\nu}$ for some constants $C, r, \nu > 0$. Our results extend and sharpen these findings in two directions. First, we show that the same Lyapunov function can be used on $\mathbb{R}^d$, and that $\beta < (\gamma - 2)/\gamma$ is both sufficient and necessary for uniform ergodicity. Second, sampling algorithms derived from LTD need discretization and Metropolis adjustment. In order to avoid low acceptance rates in the tails, the discrete-time Metropolized samplers require carefully designed state-dependent proposals or computationally intensive discretization methods. Otherwise, a naive implementation may fail to be uniformly or geometrically ergodic [25, 20]. In contrast, we directly establish uniform ergodicity of the Metropolis-adjusted chain $(Y_t)_{t \geq 0}$.

Another interesting comparison is with the stereographic projection sampler (SPS) proposed by Yang et al. [27]. When Π has polynomial tails of order γ , SPS is uniformly ergodic if and only if $\gamma \geq 2d$, and a variant of SPS based on sub-Cauchy projection is uniformly ergodic if $\gamma \geq d + 1$ [8]. The key innovation of SPS is to use stereographic projection to transform $\mathbb{R}^d$ to a pre-compact space so that a random walk proposal can move between any two states within finitely many steps. Intuitively, SPS achieves uniform ergodicity by “compressing the space,” whereas LTD and our importance-tempered RWM do so by “compressing the time.” Several other approaches have also been proposed for improving the convergence rates of RWM with heavy-tailed targets, such as heavy-tailed proposal distributions [9] and smooth variable transformations [12].

More broadly, our work belongs to a class of methods where, though one could run MCMC directly on the target Π , one instead samples from a modified stationary distribution and applies importance weighting. The discussion paper of Jennison [11] appeared to be the first to demonstrate potential advantages of importance-tempered RWM, and Gramacy et al. [7] extended it in the context of simulated tempering. Other examples include the dynamic weighting method [15, 14] and informed MCMC samplers [28, 23, 29, 13]. These methods have high acceptance probabilities and can even be entirely rejection-free. They can also be viewed as Markov chain importance sampling schemes where the auxiliary stationary distributions are constructed only implicitly. On the theoretical side, Livingstone et al. [16] recently built a general theory for continuous-time Markov chains arising from such informed MCMC methods and proved a uniform ergodicity result for discrete targets with super-exponential tails. Their continuous-time representation is of the same type as our chain $(Y_t)_{t \geq 0}$, but they consider locally balanced rather than random-walk proposals.

2 Analysis of importance-tempered RWM

2.1 Theoretical set-up

Let $(X_i)_{i \geq 1}$ be the RWM chain given in Algorithm 1 with stationary density π_β . Denote its transition kernel by $\mathcal{T}$. Assuming that the chain is ergodic, the SNIS estimator $\tilde{\Pi}_{\beta,n}(f)$ defined in equation (1) is asymptotically unbiased, since both the numerator and denominator are asymptotically unbiased by the law of large numbers for Markov chains.

To study the asymptotic variance of $\tilde{\Pi}_{\beta,n}(f)$, we introduce a continuous-time Markov chain (i.e., a pure-jump Markov process) $(Y_t)_{t \geq 0}$ with rate kernel

$$\mathcal{K}(x, B) = w(x)^{-1} \mathcal{T}(x, B), \quad \text{for } x \in \mathbb{R}^d \text{ and Borel } B \subset \mathbb{R}^d, \quad (2)$$

where we recall $w(x) = \pi(x)/\pi_\beta(x)$. In words, $(Y_t)_{t \geq 0}$ is obtained from $(X_i)_{i \geq 1}$ by inserting independent exponential holding times: at state X_i , the chain waits for a holding time $W_i \sim \text{Exp}\{w(X_i)^{-1}\}$ before moving to X_{i+1} . Therefore, the mean holding time at X_i is $w(X_i)$, and its stationary average equals one because $\int w(x) \pi_\beta(x) dx = \int \pi(x) dx = 1$. Assuming $Y_0 = X_1$, the time average of f over the trajectory of $(Y_t)_{t \geq 0}$ up to the $(n+1)$ -th jump can be written as

$$\frac{1}{\sum_{i=1}^n W_i} \int_0^{\sum_{i=1}^n W_i} f(Y_t) dt = \frac{\sum_{i=1}^n f(X_i) W_i}{\sum_{i=1}^n W_i} =: \check{\Pi}_{\beta,n}(f).$$

The only difference between $\tilde{\Pi}_{\beta,n}(f)$ and $\check{\Pi}_{\beta,n}(f)$ defined in (1) is that the importance weights are replaced by unbiased, exponentially distributed estimates. A routine comparison argument shows that the asymptotic variance of $\check{\Pi}_{\beta,n}(f)$ is at least as large as that of $\tilde{\Pi}_{\beta,n}(f)$, and thus the convergence rate of $(Y_t)_{t \geq 0}$ can be used as a proxy for the efficiency of $\tilde{\Pi}_{\beta,n}(f)$. This approach was introduced by Zanella and Roberts [28, Lemma 2] and also used in Livingstone et al. [16].

2.2 Uniform ergodicity

We recall the definition of geometric and uniform ergodicity for the chain $(Y_t)_{t \geq 0}$. We say it is geometrically ergodic (or exponentially ergodic), if $d_{\text{TV}}(\text{Law}(Y_t | Y_0 = x), \Pi) \leq G(x)\theta^t$ for some constant $\theta \in (0, 1)$ and a measurable function $G : \mathbb{R}^d \rightarrow [0, \infty)$, where d_{TV} is the total variation distance. If $\sup_{x \in \mathbb{R}^d} G(x) < \infty$, we say $(Y_t)_{t \geq 0}$ is uniformly ergodic. For our ergodicity analysis of $(Y_t)_{t \geq 0}$, we make the following assumptions on π and κ .

Assumption 1. The target density π can be expressed as $\pi(x) = \exp[\psi(\|x\|)]$ for $x \in \mathbb{R}^d$, where $\psi : [0, \infty) \rightarrow \mathbb{R}$ is a twice continuously differentiable function such that

$$\lim_{u \rightarrow \infty} u \psi'(u) = -\gamma, \text{ and } \limsup_{u \rightarrow \infty} |u^2 \psi''(u)| < K\gamma/2, \quad (3)$$

for some constants $\gamma > d$, and $K > 0$.

Assumption 2. The proposal density κ is symmetric, and $\inf_{z: \|z\| \leq M} \kappa(z) > 0$ for some $M > 0$. Further, if random vector Z has density $\kappa(z)$, then $\mathbb{E}[ZZ^\top] = I_d$ and $\mathbb{E}[\|Z\|^4] < \infty$.

Assumption 1 implies that π has polynomial tail decay of asymptotic order γ . For example, the d -dimensional t -distribution with m degrees of freedom satisfies (3) with $\gamma = m + d$. This set-up has also been considered in, e.g., Jarner and Roberts [9]. By the classical Karamata's theorem [19], π^β is integrable if $\beta > d/\gamma$, and $\int \pi(x)^\beta dx = \infty$ if $0 < \beta < d/\gamma$. When $\beta = d/\gamma$, condition (3) alone does not determine whether π^β is integrable. Assumption 2 is standard and usually satisfied in practice.

Theorem 1. *Let $(Y_t)_{t \geq 0}$ be the continuous-time Markov chain with rate kernel given by (2), and suppose that Assumptions 1 and 2 hold with $\gamma > d + 2$. If $d/\gamma < \beta < (\gamma - 2)/\gamma$, then $(Y_t)_{t \geq 0}$ is uniformly ergodic. Conversely, if in addition $\pi(x) \geq c_L \|x\|^{-\gamma}$ for some universal constant $c_L > 0$, then $(Y_t)_{t \geq 0}$ is uniformly ergodic only if $d/\gamma < \beta < (\gamma - 2)/\gamma$.*

Proof. See Section A.2 in the supplement. $\square$

The sufficiency part of Theorem 1 is proved by establishing a drift condition for $(Y_t)_{t \geq 0}$ with Lyapunov function $V(x) = C - (\|x\| \vee r)^{-\nu}$ for some $C, r, \nu > 0$. See Proposition 1 in the supplement. The same Lyapunov function was also used by Stramer and Tweedie [24] to prove uniform ergodicity of a Langevin-type diffusion on $\mathbb{R}$ with dynamics similar to $(Y_t)_{t \geq 0}$. Their analysis, however, is significantly simpler due to the one-dimensional setting and the absence of Metropolis adjustment. Our proof requires second-order Taylor expansions together with careful control of the remainder terms. Thanks to the scaling factor $w(x)^{-1}$ of the kernel $\mathcal{K}$, we only need to prove that the RWM kernel $\mathcal{T}$ satisfies

$$(\mathcal{T}V)(x) - V(x) \leq -\tilde{\alpha}(x)V(x), \text{ whenever } \|x\| \text{ is sufficiently large,}$$

where $\tilde{\alpha}(x) > 0$ may go to zero as $\|x\| \rightarrow \infty$ but at a rate slower than $\pi(x)^{1-\beta}$. This is the key difference from most existing drift-and-minorization analysis of symmetric Metropolis–Hastings algorithms [10, 26, 4]. The necessity part of Theorem 1 is proved by establishing another drift condition with test function $\tilde{V}(x) = \log(1 + \|x\|^2)$.

Uniform ergodicity yields some practical benefits for Algorithm 1. In particular, the drift condition established in Proposition 1 implies, by Down et al. [5, Thm. 6.1], that

$$\sup_x \mathbb{E}_x[e^{\alpha_R \tau_R}] \leq 2, \text{ and thus } \sup_x \mathbb{E}_x[\tau_R] \leq \alpha_R^{-1},$$

where τ_R denotes the first time that $(Y_t)_{t \geq 0}$ enters the ball $\mathcal{B}_R = \{x: \|x\| \leq R\}$, and $\mathbb{E}_x$ denotes the probability measure under which $Y_0 = x$ a.s. That is, regardless of the initialization, the time it takes $(Y_t)_{t \geq 0}$ to enter $\mathcal{B}_R$ has a uniform exponential tail bound. For the estimator $\tilde{\Pi}_{\beta,n}(f)$, this means that the importance weight accumulated in low-density regions is uniformly controlled. However, this does not imply that the jump count

n (i.e., number of RWM iterations) is controlled. With poor initialization, the number of jumps needed to enter $\mathcal{B}_R$ could be very large.

A simple comparison argument shows that $(Y_t)_{t \geq 0}$ is always uniformly ergodic when Π has exponentially light tails. In contrast, standard RWM is only geometrically ergodic in this case.

Corollary 1. *Let $(Y_t)_{t \geq 0}$ be the continuous-time Markov chain with rate kernel given by (2). Suppose that Assumption 2 holds, and $\pi(x) \propto \exp(-\|x\|^a)$ for some $a > 0$. Then $(Y_t)_{t \geq 0}$ is uniformly ergodic for any $\beta \in (0, 1)$.*

Proof. See Section A.3 in the supplement. $\square$

2.3 Spectral gap bounds

We now turn to quantitative convergence through spectral gap bounds. For Metropolis–Hastings samplers on $\mathbb{R}^d$, explicit lower bounds on the global spectral gap are typically difficult to obtain because the state space is unbounded and the tail region can be a bottleneck. Even for RWM with log-concave targets, such bounds have only been obtained recently [1]. However, the uniform ergodicity result suggests that the tail region is not a bottleneck for our continuous-time chain $(Y_t)_{t \geq 0}$. In this section, we rigorously show that the global spectral gap of $(Y_t)_{t \geq 0}$ is indeed mainly driven by its behavior inside a sufficiently large ball, while the contribution from the tails can be controlled via the same drift condition used to prove uniform ergodicity. To formulate this localization result in full generality, we first recall the definitions of the Dirichlet form and spectral gap associated with a general non-negative symmetric kernel.

Definition 1. Let $\mathcal{Q}$ be a non-negative transition kernel on $\mathbb{R}^d$ such that $\mathcal{Q}(x, \mathbb{R}^d) < \infty$ for each $x \in \mathbb{R}^d$, and $\int_A \mathcal{Q}(x, B) \Pi(\mathrm{d}x) = \int_B \mathcal{Q}(x, A) \Pi(\mathrm{d}x)$ for any Borel $A, B \subset \mathbb{R}^d$. The Dirichlet form and spectral gap of $\mathcal{Q}$ are defined by

$$\begin{aligned} \mathcal{E}_{\mathcal{Q}}(f, f) &= \frac{1}{2} \int_{\mathbb{R}^d \times \mathbb{R}^d} [f(y) - f(x)]^2 \mathcal{Q}(x, \mathrm{d}y) \Pi(\mathrm{d}x), \\ \text{Gap}(\mathcal{Q}) &= \inf_{f: \text{Var}_{\Pi}(f) > 0} \frac{\mathcal{E}_{\mathcal{Q}}(f, f)}{\text{Var}_{\Pi}(f)}, \end{aligned}$$

where $\text{Var}_{\Pi}(f) = \int_{\mathbb{R}^d} [f(x) - \Pi(f)]^2 \Pi(\mathrm{d}x)$.

The kernel $\mathcal{Q}$ in Definition 1 can be either the rate kernel of a continuous-time Markov chain or the transition kernel of a discrete-time chain. The symmetric condition implies that the resulting Markov chain is reversible with respect to Π . We now prove a general theorem that uses the local spectral gap to control the global one; it can be viewed as an extension of the well-known result of Bakry et al. [2, Thm. 1.4] for the Langevin diffusion to Markov chains.

Theorem 2. Let $\mathcal{Q}$ be as given in Definition 1. For $R > 0$, define

$$\mathcal{B}_R = \{x \in \mathbb{R}^d : \|x\| \leq R\}, \quad p_R = \Pi(\mathcal{B}_R), \quad \Pi_R(A) = \frac{\Pi(A \cap \mathcal{B}_R)}{p_R}.$$

Let $\mathcal{Q}_R$ denote the restriction of $\mathcal{Q}$ to $\mathcal{B}_R$, which is obtained by rejecting jumps outside of $\mathcal{B}_R$:

$$\mathcal{Q}_R(x, A) = \mathcal{Q}(x, A) + \mathcal{Q}(x, \mathcal{B}_R^c)1_A(x), \text{ for } x \in \mathcal{B}_R, \text{ and Borel } A \subset \mathcal{B}_R.$$

Suppose that for some function $V: \mathbb{R}^d \rightarrow (0, \infty)$ and constants $\alpha_R, b_R > 0$, we have

$$\int_{\mathbb{R}^d} \left[\frac{V(y)}{V(x)} - 1 \right] \mathcal{Q}(x, dy) \leq -\alpha_R 1_{\mathcal{B}_R^c}(x) + b_R 1_{\mathcal{B}_R}(x).$$

If $2b_R(1 - p_R) \leq \alpha_R p_R$, then

$$\text{Gap}(\mathcal{Q}) \geq \lambda_R \left[1 + \frac{2(\lambda_R + b_R)}{\alpha_R p_R} \right]^{-1},$$

where $\lambda_R = \text{Gap}(\mathcal{Q}_R)$ is the spectral gap of $\mathcal{Q}_R$ with respect to the stationary distribution Π_R .

Proof. See Section A.4 in the supplement. $\square$

We apply Theorem 2 to $(Y_t)_{t \geq 0}$. To keep the presentation concrete, we focus on multivariate t targets, for which the constants p_R, α_R, b_R can be explicitly bounded.

Theorem 3. Consider the continuous-time Markov chain $(Y_t)_{t \geq 0}$ with rate kernel $\mathcal{K}$ given by (2). Let Π be the multivariate t -distribution with m degrees of freedom and density

$$\pi(x) = \frac{\Gamma[(m+d)/2]}{\Gamma(m/2)(m\pi)^{d/2}} \left(1 + \frac{\|x\|^2}{m} \right)^{-(m+d)/2}.$$

Assume $m > 2 + 2\epsilon$ for some $\epsilon \in (0, 1/2)$. Set $\gamma = m + d$ and choose $\beta = (\gamma - 2 - \epsilon)/\gamma$. Assume $\sigma \leq \gamma^2$ and that the proposal density κ satisfies Assumption 2 with $\int \|z\|^4 \kappa(z) dz \leq \tilde{M}d^2$ for some universal constant $\tilde{M} > 0$. Then,

$$\text{Gap}(\mathcal{K}) \geq \frac{\sigma^2 \epsilon}{65R^2} \text{Gap}(\mathcal{K}_R), \text{ for any } R \geq \max \left\{ 2\gamma, \left(\frac{32(e\gamma)^{m/2}}{\sigma^2 \epsilon} \right)^{\frac{1}{m-2}}, \frac{C\sigma\gamma^2 d^2}{\epsilon} \right\},$$

where $C \in (0, \infty)$ is a sufficiently large universal constant.

Proof. See Section A.5 in the supplement. $\square$

Remark 1. Theorem 3 provides a non-asymptotic bound on the spectral gap of $(Y_t)_{t \geq 0}$, where σ, m, ϵ are not assumed to be fixed constants. If m is fixed and $\sigma^2 = d^{-\eta}$ for some $\eta \in \mathbb{R}$ (which is commonly used in practice), then γ is of the same order as d , and

$$R = O(d^a), \text{ where } a = \max \left\{ \frac{m+2\eta}{2(m-2)}, 4 - \frac{\eta}{2} \right\}.$$

So we have effectively localized a global spectral-gap problem on $\mathbb{R}^d$ to a ball with radius polynomial in d . Particularly, if $\text{Gap}(\mathcal{K}_R)$ scales polynomially in d , so does $\text{Gap}(\mathcal{K})$. For comparison, standard RWM is not geometrically ergodic with polynomial-tailed targets and thus does not admit a positive spectral gap. An explicit lower bound on $\text{Gap}(\mathcal{K}_R)$ is left for future work.

3 Concluding remarks

Our theory characterizes the convergence of the chain $(Y_t)_{t \geq 0}$ in continuous time t , rather than the performance of Algorithm 1 as a function of the iteration count n . Therefore, in Section B of the supplement, we present simulation studies for both $(Y_t)_{t \geq 0}$ and Algorithm 1 under different initialization schemes. We find that, with a suitable choice of β , importance-tempered RWM is indeed substantially more efficient and more robust to initialization than standard RWM. Though the tempered chain tends to spend many more iterations in the tails especially under poor initialization, the resulting SNIS estimator can have much smaller asymptotic variance regardless of initialization, illustrating that uniform ergodicity effectively controls the tail contribution. We also find that it is generally preferable to choose β slightly below $(\gamma - 2)/\gamma$, the theoretical threshold for uniform ergodicity of $(Y_t)_{t \geq 0}$.

Although this paper focuses on RWM, the same importance-tempering idea is much more general. One can run essentially any MCMC sampler against the tempered target and then perform importance weighting, typically at almost no additional implementation cost.

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Supplementary material

The Supplementary Material includes proofs and simulation studies.

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Supplement to “Uniform ergodicity and spectral gap of importance-tempered MCMC with heavy-tailed targets”

By Quan Zhou

A Proofs

A.1 Drift condition for the continuous-time chain

We first prove a drift condition for the continuous-time chain $(Y_t)_{t \geq 0}$, which can be described by its generator $\mathcal{A}$ [6, Chap. 4.2]:

$$\begin{aligned} (\mathcal{A}g)(x) &= \int_{\mathbb{R}^d} [g(y) - g(x)] \mathcal{K}(x, dy) \\ &= \frac{1}{w(x)} \int_{\mathbb{R}^d} [g(y) - g(x)] \mathcal{T}(x, dy) \\ &= \frac{1}{w(x)} \int_{\mathbb{R}^d} \kappa(z) [g(x + \sigma z) - g(x)] \left[1 \wedge \frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} \right] dz, \end{aligned} \quad (\text{A4})$$

where we have used that $\mathcal{T}$ is the RWM kernel associated with Algorithm 1.

Proposition 1. *Let $\mathcal{A}$ be given by (A4), and for $R > 0$, define*

$$V(x) = 1 + (R/2)^{-\nu} - \max\{\|x\|, R/2\}^{-\nu}, \text{ where } \nu = \frac{1}{2} \min\{1, \gamma(1 - \beta) - 2\}, \quad (\text{A5})$$

Suppose Assumptions 1 and 2 hold. If $\gamma > d + 2$ and $d/\gamma < \beta < (\gamma - 2)/\gamma$, then for all sufficiently large R , there exists a finite constant $\alpha_R > 0$ such that

$$(\mathcal{A}V)(x) \leq -\alpha_R V(x), \text{ whenever } \|x\| > R. \quad (\text{A6})$$

Further, $\alpha_R \rightarrow \infty$ as $R \rightarrow \infty$.

Proof of Proposition 1. Let $R_0 \geq 2$ be the smallest value such that

$$u \psi'(u) < -\gamma + 1, \quad |u \psi'(u)| < 2\gamma, \quad |u^2 \psi''(u)| < K\gamma, \text{ for any } u \geq R_0/2. \quad (\text{A7})$$

Assumption 1 guarantees that R_0 is finite. Let K be given by (3). We will prove that it suffices to choose

$$R \geq \max \left\{ R_0, \frac{\sigma}{\nu} (C\sigma + K'\gamma^2)(1 + \mathbb{E}\|Z\|^4) \right\}, \quad (\text{A8})$$

where $C = 94$, $K' = 4(K + 6)^2$ and Z is a random vector with density κ . We note that the conditions $\gamma > d + 2$, $d/\gamma < \beta < (\gamma - 2)/\gamma$ and $R_0 \geq 2$ imply that

$$\begin{aligned} 0 < \nu \leq \frac{1}{2}, \quad 1 \leq V(x) < 2, \quad \psi'(u) < 0 \text{ for any } u \geq R_0/2, \\ \text{and } Z_\beta := \int_{\mathbb{R}^d} \pi(x)^\beta dx < \infty. \end{aligned}$$

We claim that the drift condition (A6) holds for any

$$\alpha_R \leq \inf_{x: \|x\| > R} \frac{\sigma^2 \nu}{2 \|x\|^{\nu+2} Z_\beta \pi(x)^{1-\beta}}. \quad (\text{A9})$$

Assumption 1 implies that $\pi(x)$ is regularly varying with index γ . Thus, for some slowly varying function L [19],

$$\pi(x) \propto \|x\|^{-\gamma} L(\|x\|).$$

Define

$$\delta = \gamma(1 - \beta) - 2 - \nu \geq \frac{\gamma(1 - \beta) - 2}{2} > 0,$$

where the first inequality follows from the definition of ν , and the second inequality follows from the condition $\beta < (\gamma - 2)/\gamma$. Then,

$$\|x\|^{\nu+2} \pi(x)^{1-\beta} \propto \|x\|^{\nu+2-\gamma(1-\beta)} L(\|x\|)^{1-\beta} \leq \|x\|^{-\delta} L(\|x\|)^{1-\beta},$$

which is bounded from above and tends to zero as $\|x\| \rightarrow \infty$. This proves that we can pick $\alpha_R > 0$ such that $\alpha_R \uparrow \infty$ as $R \uparrow \infty$.

To prove the drift condition (A6), it suffices to establish the following drift condition for the Markov transition kernel $\mathcal{T}$:

$$(\mathcal{T}V)(x) := \int_{\mathbb{R}^d} V(y) \mathcal{T}(x, dy) \leq \left[1 - \frac{\sigma^2 \nu}{2 \|x\|^{\nu+2}} \right] V(x), \quad \text{whenever } \|x\| > R. \quad (\text{A10})$$

Then (A6) and the choice of α_R given by (A9) follow from the definition of $\mathcal{A}$ given in (A4) and that $w(x) = \pi(x)/\pi_\beta(x) = Z_\beta \pi(x)^{1-\beta}$. To prove (A10), fix an arbitrary x such that $\|x\| > R$. Since $\mathcal{T}$ is the transition kernel of the RWM algorithm that proposes the next state x' using $x' = x + \sigma Z$ with $Z \sim \kappa$,

$$\begin{aligned} (\mathcal{T}V)(x) - V(x) &= \int_{\mathbb{R}^d} \kappa(z) [V(x + \sigma z) - V(x)] \left[1 \wedge \frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} \right] dz \\ &= \int_{\mathbb{R}^d} \kappa(z) [V(x - \sigma z) - V(x)] \left[1 \wedge \frac{\pi(x - \sigma z)^\beta}{\pi(x)^\beta} \right] dz, \end{aligned}$$

where the second equality uses that κ is symmetric. Hence, we can equivalently express $\mathcal{T}V$ by

$$(\mathcal{T}V)(x) - V(x) = \frac{1}{2} \mathbb{E}_{Z \sim \kappa} [S(x, Z)]. \quad (\text{A11})$$

The function $S(x, z)$ is defined as

$$\begin{aligned} S(x, z) &= [V(x + \sigma z) - V(x)] \left[1 \wedge \frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} \right] + [V(x - \sigma z) - V(x)] \left[1 \wedge \frac{\pi(x - \sigma z)^\beta}{\pi(x)^\beta} \right] \\ &= S_0(x, z) + S_{1,+}(x, z) \mathbf{1}_{\{\pi(x + \sigma z) < \pi(x)\}} + S_{1,-}(x, z) \mathbf{1}_{\{\pi(x - \sigma z) < \pi(x)\}}, \end{aligned} \quad (\text{A12})$$

where

$$S_0(x, z) = [V(x + \sigma z) - V(x)] + [V(x - \sigma z) - V(x)], \quad (\text{A13})$$

$$S_{1,+}(x, z) = (V(x + \sigma z) - V(x)) \left(\frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} - 1 \right), \quad (\text{A14})$$

$$S_{1,-}(x, z) = (V(x - \sigma z) - V(x)) \left(\frac{\pi(x - \sigma z)^\beta}{\pi(x)^\beta} - 1 \right).$$

We use Taylor expansion to approximate V and π ; the detailed derivations are given in Lemma 1. The bound for $\mathbb{E}[S_0(x, Z)]$ is derived in Lemma 2, and the bound for $\mathbb{E}[S_1(x, Z)1_{\{\pi(x+\sigma Z) < \pi(x)\}}]$ is derived in Lemma 3. Due to symmetry, we then obtain from (A12) and Lemmas 2 and 3 that

$$\mathbb{E}[S(x, Z)] \leq \frac{\sigma^2 \nu}{\|x\|^{\nu+2}} [d - \nu - 2 + \beta \|x\| \psi'(\|x\|)] + \frac{\sigma^2 H(\gamma, \sigma, \nu)}{\|x\|^{\nu+3}},$$

where

$$\begin{aligned} H(\gamma, \sigma, \nu) &\leq [17 + (8 + 4C_K)\gamma^2] \sigma \mathbb{E}\|Z\|^3 + (23 + 24)\sigma^{\nu+1} \mathbb{E}\|Z\|^{\nu+3} \\ &\leq [64 + 47\sigma + (8 + 4C_K)\gamma^2] \sigma (1 + \mathbb{E}\|Z\|^4) \\ &\leq [47\sigma + 2(K + 6)^2\gamma^2] \sigma (1 + \mathbb{E}\|Z\|^4) \end{aligned}$$

where the second step follows from $\mathbb{E}\|Z\|^q \leq 1 + \mathbb{E}\|Z\|^4$ for any $q \in [0, 4]$ and $\sigma^\nu \leq 1 + \sigma$, and the last step follows from $\gamma^2 \geq 4$ and the definition of C_K in Lemma 1. By our assumption on R_0 , we have

$$d - \nu - 2 + \beta \|x\| \psi'(\|x\|) < d - \beta\gamma - \nu - (2 - \beta) < -1,$$

since $d/\gamma < \beta < 1$ and $\nu > 0$. Hence,

$$\mathbb{E}[S(x, Z)] \leq -\frac{\sigma^2}{\|x\|^{\nu+2}} \left[\nu - \frac{H(\gamma, \sigma, \nu)}{\|x\|} \right] \leq -\frac{\sigma^2 \nu}{2\|x\|^{\nu+2}},$$

where the last step follows from (A8). The drift condition (A10) then follows from (A11). $\square$

Lemma 1. Consider the proof of Proposition 1. Let $\|x\| \geq R$ and $\sigma\|z\| \leq \|x\|/2$. Then,

$$\left| V(x + \sigma z) - V(x) - \frac{\nu \sigma (x^\top z)}{\|x\|^{\nu+2}} - \frac{\nu \sigma^2}{2\|x\|^{\nu+2}} \left[\|z\|^2 - \frac{(\nu + 2)(x^\top z)^2}{\|x\|^2} \right] \right| \leq \frac{8\sigma^3 \|z\|^3}{\|x\|^{\nu+3}}. \quad (\text{A15})$$

Further, if $\pi(x + \sigma z) \leq \pi(x)$,

$$\left| \frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} - 1 - \frac{\beta \sigma (x^\top z)}{\|x\|} \psi'(\|x\|) \right| \leq \frac{C_K \gamma^2 \sigma^2 \|z\|^2}{\|x\|^2}, \quad (\text{A16})$$

where $C_K = (K^2 + 10K + 20)/2$.

Proof. Fix x with $\|x\| \geq R$ and define $\varphi(\zeta) = V(x + \zeta z)$. The assumptions on x, z imply that

$$\|x + \zeta z\| \geq \|x\|/2 \geq R/2, \text{ for } \zeta \in [0, \sigma]. \quad (\text{A17})$$

Hence, $\varphi(\zeta) = 1 + (R/2)^{-\nu} - \|x + \zeta z\|^{-\nu}$ for $\zeta \in [0, \sigma]$. By second-order Taylor expansion,

$$V(x + \sigma z) - V(x) = \varphi(\sigma) - \varphi(0) = \sigma \varphi'(0) + \frac{\sigma^2}{2} \varphi''(0) + \frac{\sigma^3}{6} \varphi'''(\zeta^*) \quad (\text{A18})$$

for some $\zeta^* \in [0, \sigma]$, where

$$\begin{aligned} \varphi'(\zeta) &= \frac{\nu[(x + \zeta z)^\top z]}{\|x + \zeta z\|^{\nu+2}}, \\ \varphi''(\zeta) &= \frac{\nu\|z\|^2}{\|x + \zeta z\|^{\nu+2}} - \frac{\nu(\nu+2)[(x + \zeta z)^\top z]^2}{\|x + \zeta z\|^{\nu+4}}, \\ \varphi'''(\zeta) &= \nu(\nu+2)(\nu+4) \frac{[(x + \zeta z)^\top z]^3}{\|x + \zeta z\|^{\nu+6}} - 3\nu(\nu+2) \frac{[(x + \zeta z)^\top z]\|z\|^2}{\|x + \zeta z\|^{\nu+4}}. \end{aligned}$$

Using the Cauchy-Schwarz inequality and $\nu \leq 1/2$, we get

$$|\varphi''(\zeta)| = \nu(\nu+2) \frac{|(x + \zeta z)^\top z|}{\|x + \zeta z\|^{\nu+4}} \left| (\nu+4) \frac{[(x + \zeta z)^\top z]^2}{\|x + \zeta z\|^2} - 3\|z\|^2 \right| \leq 3\nu(\nu+2) \frac{\|z\|^3}{\|x + \zeta z\|^{\nu+3}}.$$

Using $\nu \leq 1/2$ again, we get $3\nu(\nu+2) < 4$, and by (A17), the remainder term in (A18) can be bounded by

$$\frac{1}{6} |\varphi'''(\zeta^*)| \leq \frac{4\|z\|^3}{6 \inf_{\zeta \in [0, \sigma]} \|x + \zeta z\|^{\nu+3}} \leq \frac{2^{\nu+4}\|z\|^3}{3\|x\|^{\nu+3}} \leq \frac{8\|z\|^3}{\|x\|^{\nu+3}},$$

where the last step again follows from $\nu \leq 1/2$. This proves (A15).

An analogous calculation using first-order Taylor expansion and the Cauchy-Schwarz inequality yields

$$\begin{aligned} \left| \log \frac{\pi(x + \sigma z)}{\pi(x)} - \frac{\sigma(x^\top z)}{\|x\|} \psi'(\|x\|) \right| &\leq \frac{\sigma^2 \|z\|^2 \sup_{u \geq R/2} [|u\psi'(u)| + |u^2\psi''(u)|]}{2 \inf_{\zeta \in [0, \sigma]} \|x + \zeta z\|^2} \\ &\leq \frac{(K+2)\gamma\sigma^2\|z\|^2}{2 \inf_{\zeta \in [0, \sigma]} \|x + \zeta z\|^2} \\ &\leq 2(K+2)\gamma \frac{\sigma^2\|z\|^2}{\|x\|^2}, \end{aligned} \quad (\text{A19})$$

where the second inequality uses (A7) and the last uses (A17). By the Cauchy-Schwarz inequality,

$$\left| \frac{\sigma(x^\top z)}{\|x\|} \psi'(\|x\|) \right| \leq \frac{\sigma\|z\|}{\|x\|} \|x\| |\psi'(\|x\|)| \leq \frac{2\gamma\sigma\|z\|}{\|x\|}. \quad (\text{A20})$$

Using $\sigma\|z\| \leq \|x\|/2$, (A19) and (A20), we get

$$\left| \log \frac{\pi(x + \sigma z)}{\pi(x)} \right| \leq \frac{2\gamma\sigma\|z\|}{\|x\|} + \frac{2(K+2)\gamma\sigma^2\|z\|^2}{\|x\|^2} \leq \frac{(K+4)\gamma\sigma\|z\|}{\|x\|}.$$

Finally, using $e^y - 1 - y \leq y^2/2$ for $y \leq 0$, (A19) and (A20), we find that

$$\left| \frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} - 1 - \beta \log \frac{\pi(x + \sigma z)}{\pi(x)} \right| \leq \frac{\beta^2}{2} \left(\log \frac{\pi(x + \sigma z)}{\pi(x)} \right)^2 \leq \frac{(K+4)^2 \gamma^2 \sigma^2 \|z\|^2}{2 \|x\|^2} \quad (\text{A21})$$

Combining (A19) with (A21) and using $\gamma \geq 2$, we prove (A16). $\square$

Lemma 2. *Consider the proof of Proposition 1. For any x such that $\|x\| \geq R$,*

$$\mathbb{E}[S_0(x, Z)] \leq \frac{\sigma^2 \nu (d - \nu - 2)}{\|x\|^{\nu+2}} + \frac{17\sigma^3 \mathbb{E}\|Z\|^3 + 23\sigma^{\nu+3} \mathbb{E}\|Z\|^{\nu+3}}{\|x\|^{\nu+3}}. \quad (\text{A22})$$

Proof. By (A15), we can write $S_0(x, z) = \tilde{S}_0(x, z) + \mathcal{R}_0(x, z)$, where

$$\begin{aligned} \tilde{S}_0(x, z) &= \frac{\nu \sigma^2}{\|x\|^{\nu+2}} \left[\|z\|^2 - \frac{(\nu+2)(x^\top z)^2}{\|x\|^2} \right], \\ \text{and } |\mathcal{R}_0(x, z)| &\leq \frac{16\sigma^3 \|z\|^3}{\|x\|^{\nu+3}}, \text{ if } \sigma \|z\| \leq \|x\|/2. \end{aligned} \quad (\text{A23})$$

Hence, by letting

$$A = \{\sigma \|Z\| \leq \|x\|/2\}, \quad (\text{A24})$$

we get

$$\mathbb{E}[S_0(x, Z)] = \mathbb{E}[\tilde{S}_0(x, Z)] + \mathbb{E}[\mathcal{R}_0(x, Z)1_A] + \mathbb{E}[(S_0(x, Z) - \tilde{S}_0(x, Z))1_{A^c}].$$

A straightforward calculation yields that

$$\mathbb{E}[\tilde{S}_0(x, Z)] = \frac{\sigma^2 \nu (d - \nu - 2)}{\|x\|^{\nu+2}}. \quad (\text{A25})$$

Since $V \in [1, 2)$ and $\nu \leq 1/2$, by Markov's inequality, we have

$$\mathbb{E}[|S_0(x, Z)|1_{A^c}] \leq 2\mathbb{P}(\sigma \|Z\| > \|x\|/2) \leq \frac{2(2\sigma)^{\nu+3}}{\|x\|^{\nu+3}} \mathbb{E}\|Z\|^{\nu+3} \leq \frac{23\sigma^{\nu+3}}{\|x\|^{\nu+3}} \mathbb{E}\|Z\|^{\nu+3}. \quad (\text{A26})$$

For $\tilde{S}_0(x, Z)1_{A^c}$, we first use the Cauchy-Schwarz inequality to get

$$|\tilde{S}_0(x, z)| \leq \frac{\nu(\nu+1)\sigma^2 \|z\|^2}{\|x\|^{\nu+2}}.$$

Since on A^c , $\|Z\|^2 \leq \sigma \|Z\|^3 / (\|x\|/2)$, we get

$$\mathbb{E}[|\tilde{S}_0(x, Z)|1_{A^c}] \leq \frac{\nu(\nu+1)\sigma^2}{\|x\|^{\nu+2}} \mathbb{E} \left[\frac{\sigma \|Z\|^3}{\|x\|/2} \right] = \frac{\nu(\nu+1)\sigma^3}{\|x\|^{\nu+3}} \mathbb{E}\|Z\|^3 \leq \frac{\sigma^3 \mathbb{E}\|Z\|^3}{\|x\|^{\nu+3}}. \quad (\text{A27})$$

Combining (A23), (A25), (A26) and (A27), we obtain (A22). $\square$

Lemma 3. *Consider the proof of Proposition 1. For any x such that $\|x\| \geq R$,*

$$\begin{aligned} \mathbb{E}[S_{1,+}(x, z)1_{\{\pi(x+\sigma Z) < \pi(x)\}}] &\leq \frac{\beta \nu \sigma^2}{2\|x\|^{\nu+1}} \psi'(\|x\|) \\ &\quad + \frac{(4 + 2C_K)\gamma^2 \sigma^3 \mathbb{E}\|Z\|^3 + 12\sigma^{\nu+3} \mathbb{E}\|Z\|^{\nu+3}}{\|x\|^{\nu+3}}, \end{aligned}$$

where C_K is defined in Lemma 1.

Proof. By only considering the first-order expansion, we obtain from (A15) that

$$\left| V(x + \sigma z) - V(x) - \frac{\nu \sigma (x^\top z)}{\|x\|^{\nu+2}} \right| \leq \frac{\nu(\nu+1)2^{\nu+2}\sigma^2\|z\|^2}{2\|x\|^{\nu+2}} \leq \frac{C_\nu \sigma^2\|z\|^2}{\|x\|^{\nu+2}},$$

with $C_\nu = 2.2$. Combining it with (A16) and using $\gamma \geq 2$, we get

$$\begin{aligned} S_{1,+}(x, z) &= \tilde{S}_{1,+}(x, z) + \mathcal{R}_{1,+}(x, z), \\ \text{where } \tilde{S}_{1,+}(x, z) &= \frac{\beta \nu \sigma^2 (x^\top z)^2}{\|x\|^{\nu+3}} \psi'(\|x\|), \\ \text{and } |\mathcal{R}_{1,+}(x, z)| &\leq \frac{(3 + 2C_K)\gamma^2 \sigma^3 \|z\|^3}{\|x\|^{\nu+3}}, \text{ if } \sigma\|z\| \leq \|x\|/2. \end{aligned} \quad (\text{A28})$$

Note that $\psi'(\|x\|) < 0$ by the definition of R_0 .

As in the proof of Lemma 2, we introduce the event A defined by (A24). Repeating the argument used in (A27) and using $\beta < 1, \nu \leq 1/2$, we get

$$\mathbb{E} \left[|\tilde{S}_{1,+}(x, Z)| \mathbf{1}_{A^c} \right] \leq \frac{2\gamma\beta\nu\sigma^2}{\|x\|^{\nu+2}} \mathbb{E}[\|Z\|^2 \mathbf{1}_{A^c}] \leq \frac{4\gamma\beta\nu\sigma^3}{\|x\|^{\nu+3}} \mathbb{E}\|Z\|^3 \leq \frac{2\gamma\sigma^3 \mathbb{E}\|Z\|^3}{\|x\|^{\nu+3}}. \quad (\text{A29})$$

Now define

$$E_+ = \{\pi(x + \sigma Z) < \pi(x)\}.$$

To bound $\mathbb{E}[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{E_+}]$, we approximate E_+ by the event $\{x^\top Z > 0\}$. Since $\|x + \sigma Z\| \geq R/2 \geq R_0/2$ on A , we have $\psi'(\|x + \zeta Z\|) < 0$ for any $\zeta \in [0, \sigma]$ on A . This implies that

$$E_+ \cap A = \{\|x + \sigma Z\| > \|x\|\} \cap A \supset \{x^\top Z > 0\} \cap A.$$

Using $\tilde{S}_{1,+}(x, z) < 0$, we find that

$$\begin{aligned} \mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{E_+} \right] &= \mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{E_+ \cap A} \right] + \mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{E_+ \cap A^c} \right] \\ &\leq \mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{\{x^\top Z > 0\} \cap A} \right] \\ &= \mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{\{x^\top Z > 0\}} \right] - \mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{\{x^\top Z > 0\} \cap A^c} \right] \\ &\leq \mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{\{x^\top Z > 0\}} \right] + \mathbb{E} \left[|\tilde{S}_{1,+}(x, Z)| \mathbf{1}_{A^c} \right], \end{aligned}$$

where the second term on the right-hand side is bounded by (A29). Since κ is symmetric (i.e., $\kappa(z) = \kappa(-z)$ for every z), we have

$$\mathbb{E}[(x^\top Z)^2 \mathbf{1}_{\{x^\top Z > 0\}}] = \mathbb{E}[(x^\top Z)^2 \mathbf{1}_{\{x^\top Z < 0\}}] = \frac{1}{2} \mathbb{E}[(x^\top Z)^2] = \frac{\|x\|^2}{2}.$$

It thus follows that

$$\mathbb{E} \left[\tilde{S}_{1,+}(x, Z) \mathbf{1}_{E_+} \right] \leq \frac{\beta \nu \sigma^2}{2\|x\|^{\nu+1}} \psi'(\|x\|) + \frac{2\gamma\sigma^3}{\|x\|^{\nu+3}} \mathbb{E}\|Z\|^3. \quad (\text{A30})$$

Finally, using (A28), (A30), and that $2\gamma \leq \gamma^2$, $\tilde{S}_{1,+}(x, z) < 0$, we get

$$\begin{aligned} & \mathbb{E} [S_{1,+}(x, Z)\mathbf{1}_{E_+}] \\ &= \mathbb{E} [\tilde{S}_{1,+}(x, Z)\mathbf{1}_{E_+}] + \mathbb{E} [\mathcal{R}_{1,+}(x, Z)\mathbf{1}_{E_+ \cap A}] + \mathbb{E} [(S_{1,+}(x, Z) - \tilde{S}_{1,+}(x, Z))\mathbf{1}_{E_+ \cap A^c}] \\ &\leq \frac{\beta\nu\sigma^2}{2\|x\|^{\nu+1}}\psi'(\|x\|) + \frac{(4 + 2C_K)\gamma^2\sigma^3}{\|x\|^{\nu+3}}\mathbb{E}\|Z\|^3 + \mathbb{E}[|S_{1,+}(x, Z)|\mathbf{1}_{E_+ \cap A^c}], \end{aligned}$$

where the last term can be bounded by

$$\mathbb{E}[|S_{1,+}(x, Z)|\mathbf{1}_{E_+ \cap A^c}] \leq \mathbb{P}(A^c) \leq \frac{(2\sigma)^{\nu+3}\mathbb{E}\|Z\|^{\nu+3}}{\|x\|^{\nu+3}} \leq \frac{12\sigma^{\nu+3}\mathbb{E}\|Z\|^{\nu+3}}{\|x\|^{\nu+3}},$$

since $|S_{1,+}(x, Z)| \leq 1$ on E_+ . The claim then follows. $\square$

A.2 Proof of Theorem 1

Proof. The first part of the theorem is a direct consequence of Proposition 1. Explicitly, since the function V defined in (A5) is bounded, the uniform ergodicity of $(Y_t)_{t \geq 0}$ follows from Down et al. [5, Thm. 5.2], once we can verify that the set $\mathcal{B}_R = \{x: \|x\| \leq R\}$ is petite for $(Y_t)_{t \geq 0}$. To prove $\mathcal{B}_R$ is petite, it suffices to note that (1) the average holding time at any x with $\|x\| \leq R$ is bounded by

$$0 < \inf_{x: \|x\| \leq R} Z_\beta \pi(x)^{1-\beta} \leq w(x) \leq \sup_{x: \|x\| \leq R} Z_\beta \pi(x)^{1-\beta} < \infty,$$

since ψ is continuous, and that (2) $\mathcal{B}_R$ is small for the discrete-time chain $\mathcal{T}$ under Assumptions 1 and 2.

For the other direction, we first note that if $\beta \leq d/\gamma$, then $Z_\beta = \infty$, and thus $(Y_t)_{t \geq 0}$ is not well defined. To show that $\beta < (\gamma - 2)/\gamma$ is also necessary, we first prove a drift condition for the Markov transition kernel $\mathcal{T}$ using a different test function

$$V(x) = \log(1 + \|x\|^2).$$

We still use (A11) and (A12) to bound $\mathcal{T}V - V$. Let R_0 be as defined in the proof of Proposition 1; see (A7). Fix x such that $\|x\| \geq R_0$, which implies $\psi'(\|x\|) < 0$. As in Lemma 1, by second-order Taylor expansion, we have

$$\left| V(x + \sigma z) - V(x) - \frac{2\sigma x^\top z + \sigma^2 \|z\|^2}{1 + \|x\|^2} + \frac{2\sigma^2 (x^\top z)^2}{(1 + \|x\|^2)^2} \right| \leq \frac{C\sigma^3 \|z\|^3}{\|x\|^3},$$

for any z such that $2\sigma\|z\| \leq \|x\|$, where C is a universal constant. (To simplify the notation, henceforth we use C to denote constants independent of (x, z, σ) , whose values may change from line to line.) Let $S_0(x, z), S_{1,+}(x, z)$ be as defined by (A13) and (A14), which can be approximated by

$$\begin{aligned} \tilde{S}_0(x, z) &= \frac{2\sigma^2}{1 + \|x\|^2} \left(\|z\|^2 - \frac{2(x^\top z)^2}{1 + \|x\|^2} \right) \geq -\frac{4\sigma^2 (x^\top z)^2}{(1 + \|x\|^2)\|x\|^2}, \\ \tilde{S}_{1,+}(x, z) &= \frac{2\beta\sigma^2 (x^\top z)^2}{(1 + \|x\|^2)\|x\|} \psi'(\|x\|) < 0. \end{aligned} \tag{A31}$$

When $\|z\| \leq \|x\|/(2\sigma)$, the remainder terms in both approximations can be bounded in absolute value by $C\sigma^3 \|z\|^3/\|x\|^3$ for some C that may depend on K and γ . (For this necessity, we do not need to track the dependence on γ and therefore absorb it into the constants.)

Let $\tilde{S}(x, z) = \tilde{S}_0(x, z) + \tilde{S}_{1,+}(x, z)\mathbf{1}_{\{\pi(x+\sigma Z) < \pi(x)\}} + \tilde{S}_{1,-}(x, z)\mathbf{1}_{\{\pi(x-\sigma Z) < \pi(x)\}}$ denote our approximation of $S(x, z)$. Letting $A = \{\|Z\| \leq \|x\|/(2\sigma)\}$, we claim that

$$\begin{aligned} \mathbb{E}[S(x, Z)] &= \mathbb{E}[\tilde{S}(x, Z)] + \mathbb{E}[\mathcal{R}(x, Z)\mathbf{1}_A] + \mathbb{E}[\mathcal{R}(x, Z)\mathbf{1}_{A^c}] \\ &\geq \mathbb{E}[\tilde{S}(x, Z)] - \frac{C\sigma^3 \mathbb{E}\|Z\|^3}{\|x\|^3}, \end{aligned} \tag{A32}$$

where $\mathcal{R}(x, z) = S(x, z) - \tilde{S}(x, z)$, and C is some constant that only depends on K, γ . To prove (A32), we first note that the Taylor remainder estimates guarantee that $\mathbb{E}[\mathcal{R}(x, Z)1_A]$ is of the order $\sigma^3 \mathbb{E}\|Z\|^3/\|x\|^3$. Next, we bound $|\mathcal{R}(x, z)|$ by $|S(x, Z)| + |\tilde{S}(x, Z)|$ on A^c . Since $|S(x, z)| \leq |V(x + \sigma z) - V(x)| + |V(x - \sigma z) - V(x)|$ and

$$|V(x + \sigma z) - V(x)| = \left| \log \left(1 + \frac{\|x + \sigma z\|^2 - \|x\|^2}{1 + \|x\|^2} \right) \right| \leq \left| \frac{\sigma^2 \|z\|^2 + 2\sigma x^\top z}{1 + \|x\|^2} \right| \leq 1 + \frac{2\sigma^2 \|z\|^2}{\|x\|^2},$$

we have

$$\mathbb{E}[|S(x, Z)|1_{A^c}] \leq 2\mathbb{E} \left[\frac{\|Z\|^3}{\|x\|^3/(2\sigma)^3} + \frac{2\sigma^2 \|Z\|^3}{\|x\|^3/(2\sigma)} \right] = \frac{24\sigma^3 \mathbb{E}\|Z\|^3}{\|x\|^3}.$$

An analogous argument can be used to bound $\mathbb{E}[|\tilde{S}(x, Z)|1_{A^c}]$, which proves (A32).

Although a tight bound for $\mathbb{E}[\tilde{S}(x, Z)]$ can be obtained by repeating the arguments used in Lemmas 2 and 3, in this proof we only need the following crude bound:

$$\begin{aligned} \mathbb{E}[\tilde{S}(x, Z)] &\geq \mathbb{E} \left[\tilde{S}_0(x, Z) + \tilde{S}_{1,+}(x, Z) + \tilde{S}_{1,-}(x, Z) \right] \\ &\geq -\frac{4\sigma^2}{(1 + \|x\|^2)\|x\|^2} \mathbb{E} \left[(x^\top Z)^2 + 2\gamma\beta(x^\top Z)^2 \right] \geq -\frac{4\sigma^2(1 + 2\gamma\beta)}{1 + \|x\|^2}, \end{aligned}$$

where in the second step we have used (A31) and (A7). Hence, by choosing sufficiently large R , we can ensure that

$$(\mathcal{T}V)(x) - V(x) = \frac{1}{2}\mathbb{E}[S(x, Z)] \geq -\frac{C\sigma^2}{1 + \|x\|^2}, \quad \text{if } \|x\| \geq R,$$

for some constant $C > 0$ that does not depend on x and σ . Using the definition of $\mathcal{A}$ and the tail assumption on π , we find that for $\|x\| \geq R$,

$$(\mathcal{A}V)(x) \geq -\frac{C\sigma^2}{(1 + |x|^2)Z_\beta \pi(x)^{1-\beta}} \geq -\frac{C'}{Z_\beta \|x\|^{2-(1-\beta)\gamma}}.$$

Now assume $\beta \geq (\gamma - 2)/\gamma$. Then $2 - (1 - \beta)\gamma \geq 0$, and thus there exists $\alpha > 0$ such that

$$(\mathcal{A}V)(x) \geq -\alpha, \quad \text{whenever } \|x\| \geq R.$$

If $(Y_t)_{t \geq 0}$ is uniformly ergodic, by applying Theorem 16.2.2 of Meyn and Tweedie [18] to the T -skeleton chain (i.e., the discrete-time chain $(\tilde{Y}_i)_{i \geq 1}$ with $\tilde{Y}_i = Y_{iT}$), we obtain $\sup_{x \in \mathbb{R}^d} \mathbb{E}_x[\tau_R] < \infty$. But by Dynkin's formula, given $\|x\| \geq R$,

$$\begin{aligned} \mathbb{E}_x[V(X_{t \wedge \tau_R})] &= V(x) + \mathbb{E}_x \left[\int_0^{t \wedge \tau_R} (\mathcal{A}V)(X_t) dt \right] \\ &\geq V(x) - \alpha \mathbb{E}_x[t \wedge \tau_R], \end{aligned} \tag{A33}$$

By monotone convergence theorem, $\limsup_{t \rightarrow \infty} \mathbb{E}_x[t \wedge \tau_R] = \mathbb{E}_x[\tau_R]$. Since $\mathbb{E}_x[\tau_R] < \infty$, we have $\tau_R < \infty$ a.s. under $\mathbb{P}_x$, and it follows from Fatou's lemma that

$$\begin{aligned} \limsup_{t \rightarrow \infty} \mathbb{E}_x[V(X_{t \wedge \tau_R})] &= \limsup_{t \rightarrow \infty} \mathbb{E}_x[V(X_{t \wedge \tau_R})1(\tau_R < \infty)] \\ &\leq \mathbb{E}_x \left[\limsup_{t \rightarrow \infty} V(X_{t \wedge \tau_R})1(\tau_R < \infty) \right] = \mathbb{E}_x[V(X_{\tau_R})]. \end{aligned}$$

Hence, taking $\limsup$ on both sides of (A33), we obtain that

$$\log(1 + R^2) \geq \mathbb{E}_x [V(X_{\tau_R})] \geq V(x) - \alpha \mathbb{E}_x [\tau_R].$$

However, this yields

$$\sup_{x: \|x\| \geq R} \mathbb{E}_x [\tau_R] \geq \sup_{x: \|x\| \geq R} \frac{V(x) - \log(1 + R^2)}{\alpha} = \infty,$$

which contradicts our uniform ergodic assumption on $(Y_t)_{t \geq 0}$. In other words, when $\beta \geq (\gamma - 2)/\gamma$, $(Y_t)_{t \geq 0}$ cannot be uniformly ergodic. $\square$

A.3 Proof of Corollary 1

Proof. Fix any $\beta \in (0, 1)$. Choose $\gamma > 0$ such that $d/\gamma < \beta < (\gamma - 2)/\gamma$, and define a reference probability distribution $\bar{\Pi}$ with density function

$$\bar{\pi}(x) = \exp\{\bar{\psi}(\|x\|)\} \propto (1 + \|x\|)^{-\gamma}.$$

The main idea of the proof is to show that the discrete-time drift for $(Y_t)_{t \geq 0}$ under Π is at least as strong as that under $\bar{\Pi}$, and the time-change by $w(x)^{-1}$ only strengthens this effect in the tails. So the drift condition given in Proposition 1 still holds.

Write $\pi(x) = \exp\{\psi(\|x\|)\}$ where $\psi(u) = C - u^a$ for some constant $C \in \mathbb{R}$. Then,

$$\psi'(u) = -au^{a-1}, \quad \bar{\psi}'(u) = -\frac{\gamma}{1+u}.$$

It is easy to verify that there exists some $\bar{R} < \infty$ such that for all $u \geq \bar{R}$, $\psi'(u) \leq \bar{\psi}'(u)$. Hence, for any $u \geq \bar{R}$ and $h > 0$,

$$\psi(u+h) - \psi(u) \leq \bar{\psi}(u+h) - \bar{\psi}(u).$$

Since both π and $\bar{\pi}$ are radially monotone decreasing, it follows that

$$\begin{aligned} \left[\frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} - 1 \right] \mathbf{1}_{\{\pi(x + \sigma z) < \pi(x)\}} &= \left[\frac{\pi(x + \sigma z)^\beta}{\pi(x)^\beta} - 1 \right] \mathbf{1}_{\{\bar{\pi}(x + \sigma z) < \bar{\pi}(x)\}} \\ &\leq \left[\frac{\bar{\pi}(x + \sigma z)^\beta}{\bar{\pi}(x)^\beta} - 1 \right] \mathbf{1}_{\{\bar{\pi}(x + \sigma z) < \bar{\pi}(x)\}}. \end{aligned}$$

Let V be as given in Proposition 1. By (A11) and (A12), we find that

$$(\mathcal{T}V - V)(x) \leq (\bar{\mathcal{T}}V - V)(x),$$

where $\bar{\mathcal{T}}$ denotes the RWM transition kernel with stationary density $\bar{\pi}^\beta$, i.e.,

$$(\bar{\mathcal{T}}V - V)(x) = \int_{\mathbb{R}^d} \kappa(z) [V(x + \sigma z) - V(x)] \left[1 \wedge \frac{\bar{\pi}(x + \sigma z)^\beta}{\bar{\pi}(x)^\beta} \right] dz.$$

Since $\bar{\pi}$ satisfies Assumption 1, the drift condition (A10) holds for $\bar{\mathcal{T}}$ and thus also for $\mathcal{T}$, as long as R is sufficiently large. The drift coefficient α_R given by (A9) clearly grows to infinity as R increases, since $\exp(\|x\|^a)$ grows faster than $\|x\|^{\nu+2}$ for any $\nu \geq 0$. Hence, the drift condition (A6) still holds. By Down et al. [5, Thm. 5.2], $(Y_t)_{t \geq 0}$ is uniformly ergodic. $\square$

A.4 Proof of Theorem 2

Proof. Fix some f such that $\Pi(f) = 0$. Define $J(dx, dy) := \mathcal{Q}(x, dy)\Pi(dx)$. By the reversibility assumption on $\mathcal{Q}$, J is symmetric, i.e., $J(dx, dy) = J(dy, dx)$, and it follows that

$$\begin{aligned}
& \int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{V(x) - V(y)}{V(x)} f(x)^2 J(dx, dy) \\
&= \int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{V(y) - V(x)}{V(y)} f(y)^2 J(dy, dx) \\
&= \frac{1}{2} \int \left\{ \frac{V(x) - V(y)}{V(x)} f(x)^2 + \frac{V(y) - V(x)}{V(y)} f(y)^2 \right\} J(dx, dy) \\
&\leq \frac{1}{2} \int [f(y) - f(x)]^2 J(dx, dy) \\
&= \mathcal{E}_{\mathcal{Q}}(f, f),
\end{aligned} \tag{A34}$$

where the second last step follows from the elementary inequality $af(x)^2 + a^{-1}f(y)^2 - 2f(x)f(y) = [a^{1/2}f(x) - a^{-1/2}f(y)]^2 \geq 0$. The integral on the left-hand side can be bounded by the given drift condition, which yields

$$\begin{aligned}
\int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{V(x) - V(y)}{V(x)} f(x)^2 J(dx, dy) &= \int_{\mathbb{R}^d} \left\{ - \int_{\mathbb{R}^d} \left[\frac{V(y)}{V(x)} - 1 \right] \mathcal{Q}(x, dy) \right\} f(x)^2 \Pi(dx) \\
&\geq \int_{\mathbb{R}^d} [\alpha_R \mathbf{1}_{\mathcal{B}_R^c}(x) - b_R \mathbf{1}_{\mathcal{B}_R}(x)] f(x)^2 \Pi(dx) \\
&= \alpha_R \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) - b_R \Pi(f^2 \mathbf{1}_{\mathcal{B}_R}).
\end{aligned}$$

Hence, it follows from (A34) that

$$\Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) \leq \frac{b_R}{\alpha_R} \Pi(f^2 \mathbf{1}_{\mathcal{B}_R}) + \frac{1}{\alpha_R} \mathcal{E}_{\mathcal{Q}}(f, f). \tag{A35}$$

Since $\Pi(f) = 0$, we have $\Pi(f \mathbf{1}_{\mathcal{B}_R}) = -\Pi(f \mathbf{1}_{\mathcal{B}_R^c})$. Letting $\mu_R(f) = \Pi(f \mathbf{1}_{\mathcal{B}_R})/\Pi(\mathcal{B}_R)$ denote the conditional mean under Π_R and using the Cauchy-Schwarz inequality, we find that

$$p_R^2 \mu_R(f)^2 = \Pi(f \mathbf{1}_{\mathcal{B}_R})^2 = \Pi(f \mathbf{1}_{\mathcal{B}_R^c})^2 \leq \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) \Pi(\mathcal{B}_R^c) = \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) (1 - p_R).$$

Let $\text{Var}_{\Pi_R}(f) = \int_{\mathcal{B}_R} [f(x) - \mu_R(f)]^2 \Pi_R(dx)$ denote the conditional variance under Π_R . We get

$$\begin{aligned}
\Pi(f^2 \mathbf{1}_{\mathcal{B}_R}) &= p_R [\text{Var}_{\Pi_R}(f) + \mu_R(f)^2] \leq p_R \text{Var}_{\Pi_R}(f) + \frac{1 - p_R}{p_R} \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) \\
&\leq \frac{1}{\lambda_R} \mathcal{E}_{\mathcal{Q}}(f, f) + \frac{1 - p_R}{p_R} \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}),
\end{aligned} \tag{A36}$$

The last step follows from the definition of the spectral gap λ_R

$$\lambda_R = \inf_{f: \text{Var}_{\Pi_R}(f) > 0} \frac{\mathcal{E}_R(f, f)}{\text{Var}_{\Pi_R}(f)},$$

where the Dirichlet form $\mathcal{E}_R(f, f)$ satisfies

$$\begin{aligned}\mathcal{E}_R(f, f) &= \frac{1}{2} \int_{\mathcal{B}_R \times \mathcal{B}_R} [f(y) - f(x)]^2 \Pi_R(dx) \mathcal{Q}_R(x, dy) \\ &= \frac{1}{2p_R} \int_{\mathcal{B}_R \times \mathcal{B}_R} [f(y) - f(x)]^2 \Pi(dx) \mathcal{Q}(x, dy) \leq \frac{\mathcal{E}_{\mathcal{Q}}(f, f)}{p_R}.\end{aligned}$$

Combining (A35) and (A36) and rearranging the terms, we get

$$\frac{\alpha_R p_R - b_R(1 - p_R)}{\alpha_R p_R} \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) \leq \left(\frac{b_R}{\alpha_R \lambda_R} + \frac{1}{\alpha_R} \right) \mathcal{E}_{\mathcal{Q}}(f, f). \quad (\text{A37})$$

By assumption, $\alpha_R p_R > b_R(1 - p_R)/2$. So we can now use (A36) and (A37) to get

$$\begin{aligned}\text{Var}_{\Pi}(f) &= \Pi(f^2 \mathbf{1}_{\mathcal{B}_R}) + \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) \\ &\leq \frac{1}{\lambda_R} \mathcal{E}_{\mathcal{Q}}(f, f) + \frac{1}{p_R} \Pi(f^2 \mathbf{1}_{\mathcal{B}_R^c}) \\ &\leq \frac{1}{\lambda_R} \mathcal{E}_{\mathcal{Q}}(f, f) + \frac{1 + b_R/\lambda_R}{\alpha_R p_R - b_R(1 - p_R)} \mathcal{E}_{\mathcal{Q}}(f, f) \\ &\leq \frac{\mathcal{E}_{\mathcal{Q}}(f, f)}{\lambda_R} \left[1 + \frac{2(\lambda_R + b_R)}{\alpha_R p_R} \right].\end{aligned}$$

The claim then follows from the definition of the spectral gap. □

A.5 Proof of Theorem 3

Proof. We split the proof into multiple steps.

Step 1: Drift condition on $\mathcal{B}_R^c$. Define $\gamma = m + d$, which is the tail index of Π . A simple calculation shows that $\pi(x) \propto e^{\psi(\|x\|)}$, where ψ satisfies

$$u\psi'(u) = -\frac{\gamma u^2}{u^2 + m}, \quad u^2\psi''(u) = \frac{\gamma u^2(u^2 - m)}{(u^2 + m)^2}.$$

Hence, Assumption 1 is satisfied with $K = 2$. Further, since $\gamma > d + 2$, we can define $R_0 = 2\sqrt{\gamma(\gamma - d)} \leq 2\gamma$, which satisfies $R_0 \geq 2$ and equation (A7) in the proof of Proposition 1. Hence, we can define V by

$$\begin{aligned} V(x) &= 1 + (R/2)^{-\nu} - \max\{\|x\|, R/2\}^{-\nu}, \\ \text{where } \nu &= \frac{1}{2} \min\{1, \gamma(1 - \beta) - 2\} = \frac{\epsilon}{2}, \\ R &\geq \max\left\{2\gamma, \frac{C\sigma\gamma^2 d^2}{\epsilon}\right\}, \end{aligned} \tag{A38}$$

for some universal constant $C > 0$. The condition on R follows from (A8) and the assumptions $\sigma < \gamma^2$ and $\mathbb{E}\|Z\|^4 \leq Md^2$. By (A9), the following drift condition holds:

$$\frac{(\mathcal{A}V)(x)}{V(x)} = \int_{\mathbb{R}^d} \left[\frac{V(y)}{V(x)} - 1 \right] \mathcal{K}(x, dy) \leq -\frac{\sigma^2\nu}{2\|R\|^{\nu+2} Z_\beta \pi(R)^{1-\beta}}, \text{ if } x \in \mathcal{B}_R^c.$$

Using $1 + \|x\|^2/m \geq \|x\|^2/m$, we get

$$\pi(x) \leq \frac{c_{d,m}}{\|x\|^\gamma}, \text{ where } c_{d,m} = \frac{\Gamma[(m+d)/2]m^{m/2}}{\Gamma(m/2)\pi^{d/2}}.$$

Hence, in order to ensure that $(\mathcal{A}V)(x) \leq -\alpha_R V(x)$ on $\mathcal{B}_R^c$, we can choose

$$\alpha_R = \frac{\sigma^2\nu}{2Z_\beta c_{d,m}^{1-\beta}} R^{\epsilon/2}, \tag{A39}$$

where we have used

$$\gamma(1 - \beta) - 2 - \nu = \epsilon - \nu = \frac{\epsilon}{2}.$$

Step 2: Drift condition on $\mathcal{B}_R$. Next, we find b_R such that

$$\frac{(\mathcal{A}V)(x)}{V(x)} \leq b_R, \text{ if } x \in \mathcal{B}_R.$$

This follows directly from the boundedness of V . Explicitly, since $V \in [1, 1 + (R/2)^{-\nu}]$, we have

$$\frac{(\mathcal{A}V)(x)}{V(x)} = \frac{1}{w(x)} \int_{\mathbb{R}^d} \left[\frac{V(y)}{V(x)} - 1 \right] \mathcal{T}(x, dy) \leq \frac{(R/2)^{-\nu}}{w(x)}$$

Since $R > \sqrt{\gamma m}$, we have

$$\pi(x) = \frac{c_{d,m}}{\|x\|^\gamma} \left(1 + \frac{m}{\|x\|^2}\right)^{-\gamma/2} \geq \frac{c_{d,m}}{\|x\|^\gamma} \left(1 + \frac{1}{\gamma}\right)^{-\gamma/2} \geq \frac{c_{d,m}}{\sqrt{e}\|x\|^\gamma}, \text{ whenever } \|x\| \geq R.$$

Hence, using $\gamma(1 - \beta) - \nu = 2 + \epsilon/2$, we find that the drift condition on $\mathcal{B}_R$ holds with

$$b_R = \frac{2^\nu \sqrt{e}}{Z_\beta c_{d,m}^{1-\beta}} R^{2+\epsilon/2} \geq \sup_{x: |x| \leq R} \frac{(R/2)^{-\nu}}{w(x)} = \frac{(R/2)^{-\nu}}{Z_\beta \pi(R)^{1-\beta}}. \quad (\text{A40})$$

Step 3: Bound on p_R . To bound $p_R = \Pi(\mathcal{B}_R)$, we perform the routine polar coordinate transformation, which yields

$$\begin{aligned} 1 - p_R &= \frac{\Gamma(\gamma/2)}{\Gamma(m/2)(m\pi)^{d/2}} \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_R^\infty u^{d-1} \left(1 + \frac{u^2}{m}\right)^{-\gamma/2} du \\ &\leq \frac{\Gamma(\gamma/2)}{\Gamma(m/2)(m\pi)^{d/2}} \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_R^\infty u^{d-1} \frac{u^{-\gamma}}{m^{-\gamma/2}} du \\ &= \frac{2\Gamma(\gamma/2)m^{m/2-1}}{\Gamma(m/2)\Gamma(d/2)} R^{-m} \\ &= \frac{\Gamma(\gamma/2)m^{m/2}}{\Gamma[(m+2)/2]\Gamma(d/2)} R^{-m}. \end{aligned}$$

To bound the right-hand side, we use the following standard inequalities for the gamma function [3],

$$y^{x-y} \leq \frac{\Gamma(x)}{\Gamma(y)} \leq x^{x-y}, \quad \Gamma(x+1) \geq \left(\frac{x}{e}\right)^x, \quad \text{whenever } x > y > 0. \quad (\text{A41})$$

It follows that

$$1 - p_R \leq \frac{m^{m/2} R^{-m}}{\Gamma[(m+2)/2]} \left(\frac{\gamma}{2}\right)^{m/2} \leq \frac{m^{m/2}}{R^m} \left(\frac{\gamma}{2}\right)^{m/2} \left(\frac{2e}{m}\right)^{m/2} = (\gamma e)^{m/2} R^{-m}. \quad (\text{A42})$$

Since $R \geq 2\gamma$ and $\gamma = m + d > d + 2$, one can numerically verify that $p_R \geq 1/2$.

Step 4: Bound on λ_R . We now find a simple upper bound on λ_R to show that it has a smaller order than b_R . By the definition of the spectral gap, we can pick any square-integrable test function f and use $\mathcal{E}_R(f, f)/\text{Var}_{\Pi_R}(f)$ as an upper bound, where $\mathcal{E}_R$ denotes the Dirichlet form associated with the kernel $\mathcal{K}_R$. We simply choose $f(x) = x_1$, where x_1 denotes the first coordinate of $x \in \mathbb{R}^d$. Let Z have density κ . Then,

$$\begin{aligned} \mathcal{E}_R(f, f) &= \frac{1}{2} \int_{\mathcal{B}_R \times \mathcal{B}_R} [f(y) - f(x)]^2 \mathcal{K}_R(x, dy) \Pi_R(dx) \\ &= \frac{1}{2p_R} \int_{\mathcal{B}_R \times \mathcal{B}_R} \frac{[f(y) - f(x)]^2}{w(x)} \mathcal{T}(x, dy) \Pi(dx), \end{aligned}$$

where $\mathcal{T}$ is the RWM kernel with proposal density κ . Hence, we can bound $\int_{\mathcal{B}_R} [f(y) - f(x)]^2 \mathcal{T}(x, dy)$ by assuming all proposals are accepted, which yields

$$\int_{\mathcal{B}_R} [f(y) - f(x)]^2 \mathcal{T}(x, dy) \leq \int_{\mathbb{R}^d} [f(x + \sigma z) - f(x)]^2 \kappa(z) dz = \sigma^2,$$

where the last step follows from the definition of f and Assumption 2. Since $\pi(x)/w(x) = \pi_\beta(x)$, where π_β denotes the normalized tempered density function, we obtain that

$$\mathcal{E}_R(f, f) \leq \frac{\sigma^2}{2p_R} \int_{\mathcal{B}_R} \frac{1}{w(x)} \Pi(dx) = \frac{\sigma^2}{2p_R} \int_{\mathbb{R}^d} \pi_\beta(x) dx = \frac{\sigma^2}{2p_R}. \quad (\text{A43})$$

Meanwhile, under Π , the first coordinate also follows the univariate t distribution with m degrees of freedom. Let $\pi_1(x_1) \propto (1 + m^{-1}x_1^2)^{-(m+1)/2}$ denote the corresponding density function. Then,

$$\text{Var}_{\Pi_R}(f) = \frac{\int_{-R}^R x_1^2 \pi_1(x_1) dx_1}{\int_{-R}^R \pi_1(x_1) dx_1} \geq \int_{-R}^R x_1^2 \pi_1(x_1) dx_1 = 2 \int_0^R x_1^2 \pi_1(x_1) dx_1$$

Using $m > 2$, $R > \gamma \geq m + 1$ and (A41), we bound the integral on the right-hand side by

$$\begin{aligned} \int_0^R x_1^2 \pi_1(x_1) dx_1 &\geq \frac{\Gamma[(m+1)/2]}{\sqrt{m\pi}\Gamma(m/2)} \int_1^2 x_1^2 \left(1 + \frac{x_1^2}{m}\right)^{-(m+1)/2} dx_1 \\ &\geq \frac{1}{\sqrt{2\pi}} \int_1^2 e^{-x_1^2(m+1)/(2m)} dx_1 \\ &\geq \frac{1}{\sqrt{2\pi}} \int_1^2 e^{-3x_1^2/4} dx_1 \geq \frac{e^{-3}}{\sqrt{2\pi}}. \end{aligned}$$

Combining this with (A43) and using $p_R \geq 1/2$, we conclude that $\lambda_R \leq C_\lambda \sigma^2$ for some universal constant $C_\lambda > 0$.

Step 5: Application of Theorem 2. To apply Theorem 2, we need to ensure that $2b_R(1 - p_R) \leq \alpha_R p_R$, which holds if $\alpha_R \geq 4b_R(1 - p_R)$ since $p_R \geq 1/2$. By combining (A39), (A40) and (A42), we find that this condition holds when

$$\frac{\sigma^2 \nu}{2^{\nu+3} \sqrt{e} (\gamma e)^{m/2}} R^{m-2} \geq 1.$$

Since $\nu = \epsilon/2 \in (0, 1/4)$, it suffices to require that

$$R \geq \left(\frac{32}{\sigma^2 \epsilon} \right)^{\frac{1}{m-2}} (\gamma e)^{\frac{m}{2(m-2)}}. \quad (\text{A44})$$

Combining (A38) and (A44) gives the claimed choice of R in the theorem statement.

To compare λ_R with b_R , we find an upper bound on $Z_\beta c_{d,m}^{1-\beta}$. First, a routine calculation yields

$$Z_\beta = \int \pi(x)^\beta dx = (m\pi)^{(1-\beta)d/2} \frac{\Gamma(\gamma/2)^\beta}{\Gamma(m/2)^\beta} \frac{\Gamma[(\beta\gamma - d)/2]}{\Gamma(\beta\gamma/2)}.$$

Hence,

$$\begin{aligned} Z_\beta c_{d,m}^{1-\beta} &= m^{(1-\beta)\gamma/2} \frac{\Gamma(\gamma/2)}{\Gamma(m/2)} \frac{\Gamma[(\beta\gamma - d)/2]}{\Gamma(\beta\gamma/2)} \\ &= m^{(2+\epsilon)/2} \left[\frac{\Gamma(\gamma/2)}{\Gamma(\beta\gamma/2)} \right] \left[\frac{\Gamma((\beta\gamma - d)/2)}{\Gamma((\gamma - d)/2)} \right], \end{aligned}$$

where in the second step we use $\beta = (\gamma - 2 - \epsilon)/\gamma$. By (A41),

$$Z_\beta c_{d,m}^{1-\beta} \leq m^{(2+\epsilon)/2} \left(\frac{\gamma}{2} \right)^{(1-\beta)\gamma/2} \left(\frac{\beta\gamma - d}{2} \right)^{(1-\beta)\gamma/2} = \left(\frac{m\gamma}{m - 2 - \epsilon} \right)^{(2+\epsilon)/2}.$$

Hence, for the choice of b_R given in (A40), we have

$$b_R \geq \left(\frac{m-2-\epsilon}{m\gamma} \right)^{(2+\epsilon)/2} R^{(4+\epsilon)/2} \geq \left(\frac{m-2-\epsilon}{\gamma^2} \right)^{(2+\epsilon)/2} R^2.$$

By assumption we have $m > 2 + 2\epsilon$, and by (A38) we have $R \geq C\sigma\gamma^2/\epsilon$. It follows that

$$b_R \geq \left(\frac{\epsilon}{\gamma^2} \right)^{(2+\epsilon)/2} \left(\frac{C\sigma\gamma^2}{\epsilon} \right)^2 \geq C^2\sigma^2\gamma^{2-\epsilon} \geq \gamma C^2\sigma^2,$$

since $\epsilon \in (0, 1/2)$ and $(2 + \epsilon)/2 < 2$. Since $\lambda_R \leq C\lambda\sigma^2$, choosing sufficiently large C guarantees that $b_R \geq \lambda_R$. Since $p_R \geq 1/2$, by Theorem 2, we get

$$\text{Gap}(\mathcal{K}) \geq \lambda_R \left[1 + \frac{2(\lambda_R + b_R)}{\alpha_R p_R} \right]^{-1} \geq \lambda_R \left[1 + \frac{8b_R}{\alpha_R} \right]^{-1},$$

where $\lambda_R = \text{Gap}(\mathcal{K}_R)$. Since

$$\frac{b_R}{\alpha_R} = \frac{2^{\nu+1}\sqrt{e}}{\sigma^2\nu} R^2 = \frac{2^{\nu+2}\sqrt{e}}{\sigma^2\epsilon} R^2 \leq \frac{8R^2}{\sigma^2\epsilon},$$

and $R^2/(\sigma^2\epsilon) \geq C^2\gamma^4/\epsilon^3 > 1$ whenever C is sufficiently large, we obtain the claimed bound. $\square$

B Numerical examples

B.1 Uniform ergodicity of the continuous-time chain

In the first numerical example, we let Π be the multivariate t -distribution with $m = 4$ degrees of freedom. Theorem 1 implies that $(Y_t)_{t \geq 0}$ is uniformly ergodic if and only if

$$\frac{d}{d+4} < \beta < \frac{d+2}{d+4}.$$

For comparison, we consider two choices of β : $\beta_1 = (d+2-\epsilon)/(d+4)$ and $\beta_2 = (d+2+\epsilon)/(d+4)$ with $\epsilon = 1/4$. We let the proposal density κ be the standard Gaussian distribution and tune the step size σ so that the expected acceptance rate is above 0.1 for both β_1, β_2 . Then, we simulate the continuous-time Markov chain $(Y_t)_{t \geq 0}$ with initial value $Y_0 = (y, \dots, y)$, using different values of y . We simulate 10^4 trajectories, and for each fixed t , we compare the distribution of Y_t with Π by computing Kolmogorov–Smirnov (KS) statistics for the first coordinate and for the radius $\|Y_t\|$.

In Figure 1, we visualize the resulting KS statistics on a logarithmic scale for $d = 1, 5, 10$. For each d , we consider four initialization schemes where $\|Y_0\|$ grows geometrically. It can be seen that when $\beta = (d+2-\epsilon)/(d+4)$ (i.e., $\beta = 0.55$ for $d = 1$, $\beta = 0.75$ for $d = 5$ and $\beta = 0.84$ for $d = 10$), the time needed for the KS statistic to reach a given level (say, 0.02) is not substantially affected by the starting point. This is consistent with the uniform ergodicity of $(Y_t)_{t \geq 0}$, which ensures that the time required to approach stationarity is uniformly controlled over all initial states. In contrast, when $\beta = (d+2+\epsilon)/(d+4)$ (i.e., $\beta = 0.65$ for $d = 1$, $\beta = 0.81$ for $d = 5$ and $\beta = 0.875$ for $d = 10$), the chain is not uniformly ergodic, and the dependence on initialization becomes much more pronounced. In particular, the corresponding curves are more clearly separated, and the gaps between successive curves remain roughly constant, which suggests that the time required for the KS statistic to reach a certain level will keep growing with $\|Y_0\|$.

We note that by the Dvoretzky–Kiefer–Wolfowitz bound, at significance level 0.05, the half width of the confidence interval of each KS statistic is

$$\sqrt{\frac{\log(2/0.05)}{2 \times 10^4}} \approx 0.0136.$$

This explains why the curves in Figure 1 become unstable once the KS statistic drops below 0.02, since those values are largely driven by Monte Carlo error.

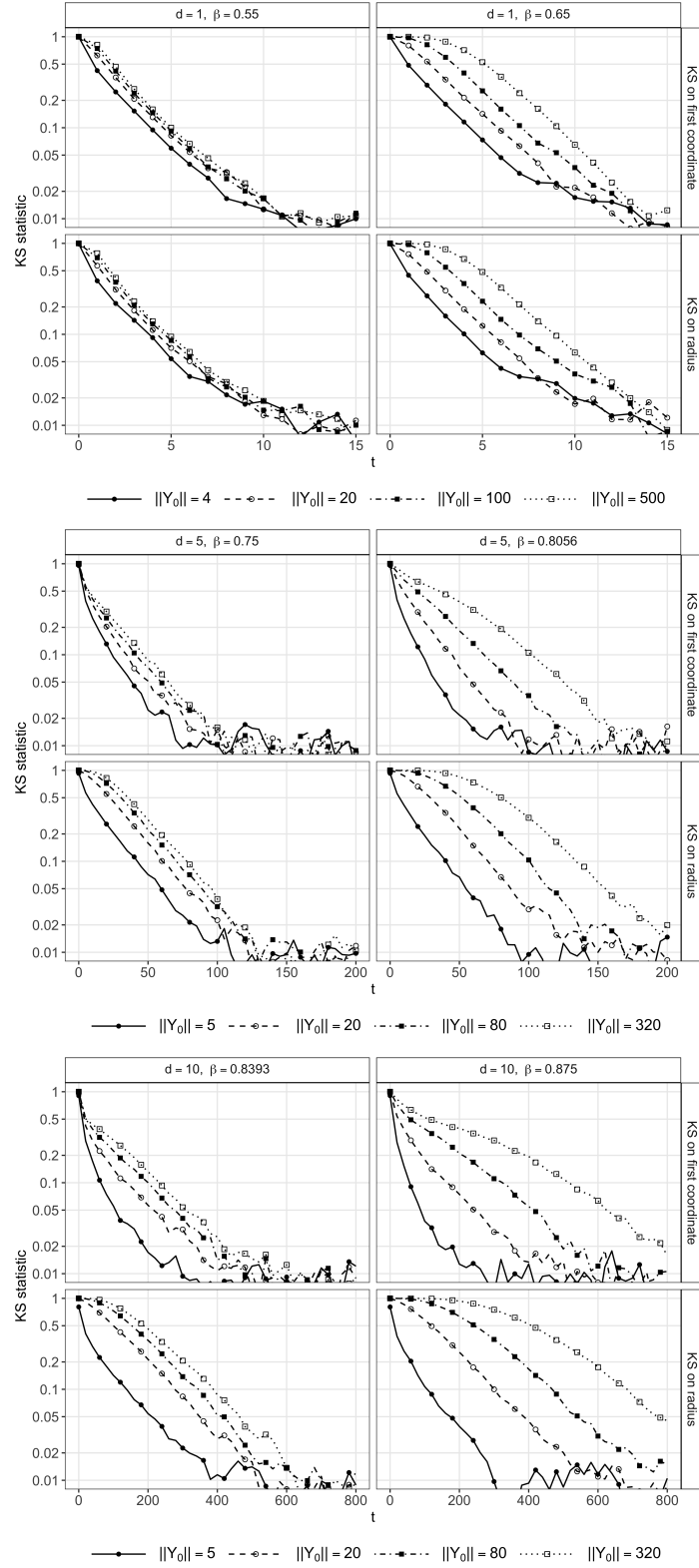


Figure 1: Convergence of the continuous-time Markov chain $(Y_t)_{t \geq 0}$ with Π being the d -dimensional t -distribution with $m = 4$ degrees of freedom. The Kolmogorov-Smirnov test statistic compares t_4 with the distribution of Y_t over 10^4 replicates. For each KS statistic, the confidence interval has width ± 0.014 .

B.2 Efficiency of importance-tempered RWM

In the second experiment, we simulate the importance-tempered RWM sampler (i.e., Algorithm 1) and numerically study the variance of the SNIS estimator $\tilde{\Pi}_{\beta,n}(f)$ defined in (1). We let Π be the multivariate t distribution with $m = 5$ degrees of freedom (so that the variance of the test function $f(x) = \|x\|^2$ is finite). Since Π_β is well defined only when $\beta > d/\gamma$ where $\gamma = d + m = d + 5$, we consider the following values of β :

$$\beta = \frac{d + 0.5k}{d + 5}, \text{ for } k = 1, 2, \dots, 10.$$

When $\beta = 1$ (i.e., $k = 10$), Algorithm 1 reduces to the usual RWM algorithm. Recall that the continuous-time chain $(Y_t)_{t \geq 0}$ is uniformly ergodic when $\beta < (d+3)/(d+5)$, and we will denote this critical value by β^* . The proposal distribution κ is Gaussian, and σ is tuned in the same way as in the first simulation study. For each choice of (d, β) , we consider three initialization schemes: (i) $X_0 \sim \Pi_\beta^{\text{trunc}}$, where Π_β^{trunc} denotes Π_β conditioned on $\|X_0\| \leq 2\sqrt{d}$, (ii) $\|X_0\| = 20$ (iii) $\|X_0\| = 40$. In the latter two schemes, X_0 is deterministic with identical coordinates.

If the sampler quickly reaches stationarity, we expect that the total importance weight $\sum_{i=1}^n w(X_i)$ is close to n , since $\int w(x)\pi_\beta(x)dx = 1$. Hence, when estimating the asymptotic variance of the SNIS estimator, we discard replicate runs for which $\sum_{i=1}^n w(X_i) < 0.05n$; in such runs, the sampler has probably not entered the high-density region, and the resulting SNIS estimates can be highly unstable. We summarize in Table 1 the number of such runs under each setting. The cutoff level $0.05n$ is chosen to be deliberately conservative. If the continuous-time chain $(Y_t)_{t \geq 0}$ is uniformly ergodic, then once the sampler reaches the high-density region, the importance weights accumulate rapidly, and the SNIS estimator stabilizes quickly thereafter.

In Figure 2, we visualize the scaled asymptotic variance for four test functions: $f(x) = x_1$, $\|x\|$, $\|x\|^2$, and $1_{\{\|x\| > q_{0.75}\}}$, where $q_{0.75}$ denotes the 75-th quantile of the radius under Π . The scaled asymptotic variance is defined as

$$\text{AV}(f) = \frac{n}{\text{Var}_\Pi(f)} \widehat{\text{Var}}\left(\tilde{\Pi}_{\beta,n}(f)\right)$$

where n is the number of iterations of Algorithm 1, and the variance of $\tilde{\Pi}_{\beta,n}(f)$ is estimated across 1,000 independent replicate runs. No burn-in samples are discarded when computing $\tilde{\Pi}_{\beta,n}(f)$. The critical value $\beta^* = (d+3)/(d+5)$ (which equals 0.667 for $d = 1$, 0.8 for $d = 5$, and 0.867 for $d = 10$) is marked on the horizontal axis. We make three important remarks.

First, in every setting, importance tempering can effectively reduce the asymptotic variance when β is appropriately chosen. The reduction becomes highly substantial when the test function is $f(x) = \|x\|^2$ or the initial state X_0 is in the tails. For the test functions $f(x) = x_1$, $\|x\|$, $1_{\{\|x\| > q_{0.75}\}}$, the optimal value of β that minimizes the asymptotic

variance seems to be close to β^* . For $f(x) = \|x\|^2$, the optimal value of β is smaller. This is because the variance of $f(x) = \|x\|^2$ is largely driven by the tail behavior (compared with the other three test functions), and a smaller β encourages the sampler to spend more time exploring tail regions.

Second, by comparing the asymptotic variances with $\beta < \beta^*$ and those with $\beta \geq \beta^*$, we see that the initialization scheme has a much more significant impact when $\beta \geq \beta^*$. This is again consistent with our theory about the chain $(Y_t)_{t \geq 0}$. When $(Y_t)_{t \geq 0}$ is uniformly ergodic (i.e., $\beta < \beta^*$), the SNIS estimator is largely insensitive to the initialization, since regardless of how poor the initialization is, the total importance weight accumulated in the tail regions is uniformly controlled. To further illustrate this phenomenon, in Figure 3 we visualize the distributions of the estimator $\tilde{\Pi}_{\beta,n}(f)$ for $f(x) = \|x\|^2$ under different initialization schemes. When $\beta < \beta^*$, the median value and distribution of the estimator remains largely the same over all 3 initialization schemes, although the case $\|X_0\| = 40$ may produce slightly more extreme estimates. When $\beta > \beta^*$, the median value clearly increases over the 3 initialization schemes, and when $\|X_0\| = 40$, the entire distribution is visibly shifted upward.

Third, we recall that Figures 2 and 3 do not include problematic MCMC runs that are indicated in Table 1. Such runs yield highly unstable estimates and thus can increase the asymptotic variance drastically. It can be seen from Table 1 that such runs typically occur when β is small, in which case the sampler has a much heavier-tailed stationary distribution and is therefore more likely to get stuck in the tails. For a fixed β , such failed runs occur more frequently when the initial state lies far in the tails. This does not contradict the uniform ergodic theory for the chain $(Y_t)_{t \geq 0}$: the uniform ergodicity only implies that the total importance weight accumulated in the tails is bounded uniformly, while the number of iterations of Algorithm 1 required to leave the tails still grows with $\|X_0\|$ and can be unbounded. Overall, this simulation study suggests a trade-off in the choice of β , and values slightly less than β^* appear to be preferable.

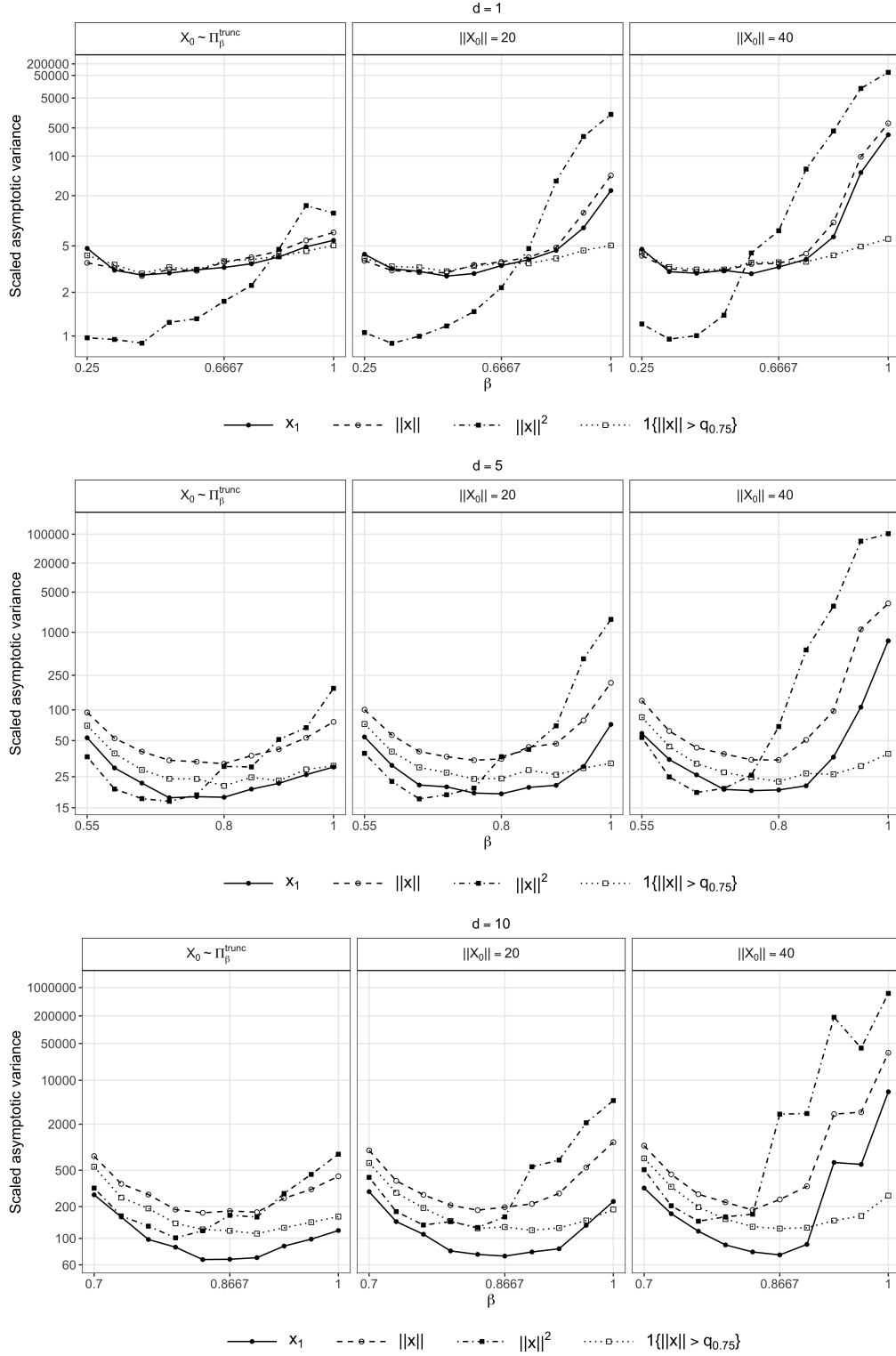


Figure 2: Simulation of the importance-tempered RWM sampler under three initialization schemes. Horizontal axis: β ; the middle tick label marks the critical value $(d+3)/(d+5)$ for uniform ergodicity of $(Y_t)_{t \geq 0}$. Vertical axis: asymptotic variance of the SNIS estimator, estimated from 1,000 replicates and scaled by $\text{Var}_\Pi(f)$. Number of iterations $n = 5,000$ for $d = 1$, $n = 40,000$ for $d = 5$, and $n = 10^5$ for $d = 10$.

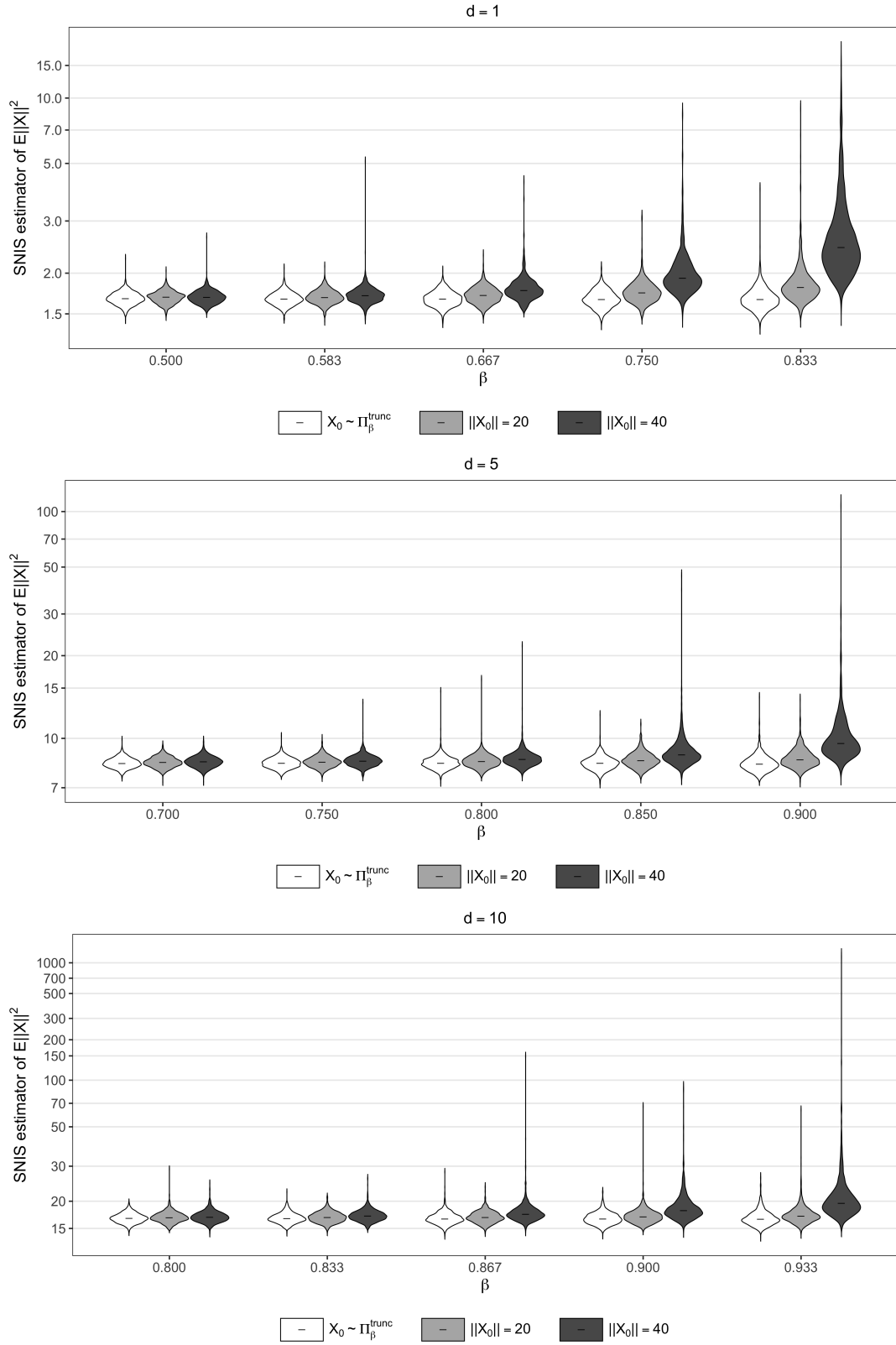


Figure 3: Distribution of the SNIS estimator for $f(x) = \|x\|^2$ over 1,000 runs of the importance-tempered RWM sampler. For each d , we consider 5 values of β , with the middle one being the critical value $(d + 3)/(d + 5)$.

Table 1: Number of removed replicates with total importance weight less than $0.05n$ out of 1,000 replicate runs. Columns corresponding to $\beta > (d + 3)/(d + 5)$ are omitted because the removal counts are all zero.

d	initialization	$\frac{d+0.5}{d+5}$	$\frac{d+1}{d+5}$	$\frac{d+1.5}{d+5}$	$\frac{d+2}{d+5}$	$\frac{d+2.5}{d+5}$	$\frac{d+3}{d+5}$
1	Π_β^{trunc}	0	0	0	0	0	0
	$\ X_0\ = 20$	5	1	0	1	0	0
	$\ X_0\ = 40$	4	1	0	0	0	0
5	Π_β^{trunc}	1	0	0	0	0	0
	$\ X_0\ = 20$	6	1	0	0	0	0
	$\ X_0\ = 40$	13	6	0	1	0	0
10	Π_β^{trunc}	1	1	0	0	0	0
	$\ X_0\ = 20$	5	3	0	1	1	0
	$\ X_0\ = 40$	32	19	4	1	1	0